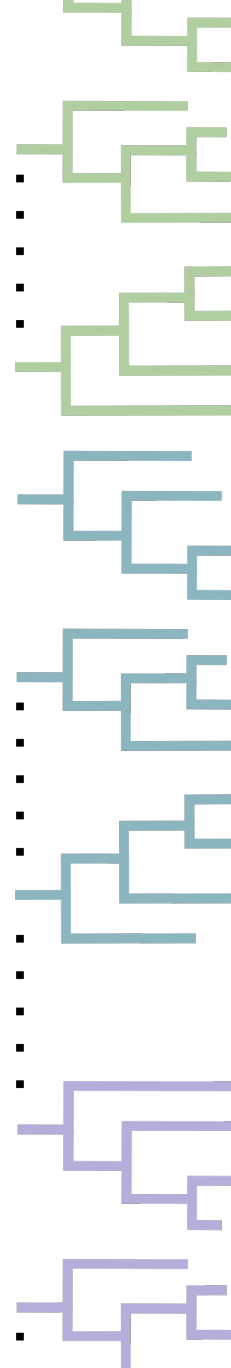


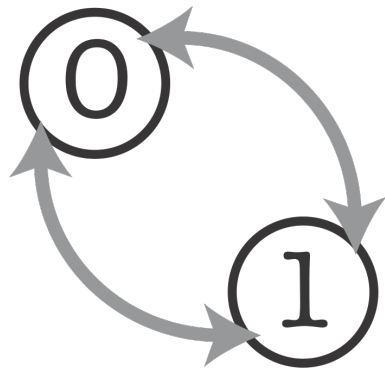
Morphological models and how to choose one

Slides: Laura Mulvey

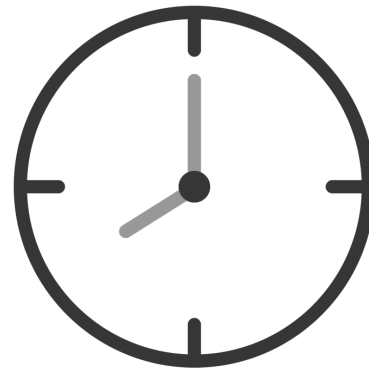


```
101510010?00-100--0000
000500010?200100--0010
102500010?200100--0?10
00?5?0010?200100?-0??1
0015000101201000430101
0015000101201010440111
??050?????201000440?11
0015000101201000430101
0015000101201010440111
??050?????201000440?11
```

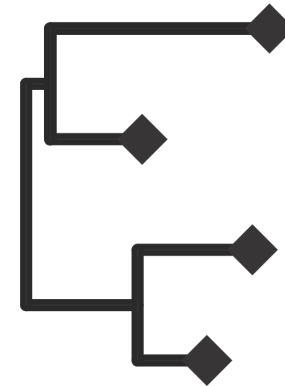
substitution model



clock model



tree model



Substitution models

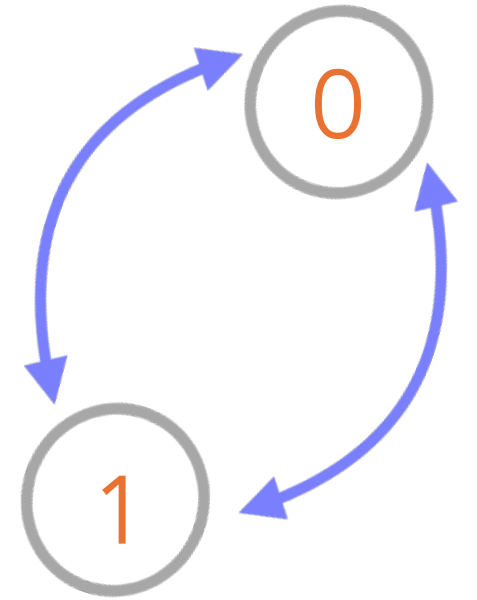
A **substitution model** is a mathematical description of how characters change over evolutionary time:

- DNA,
- RNA,
- Amino acids,
- Morphology

Mutation = when one state changes to another (e.g., $A \rightarrow G$).

Substitution = when that mutation *sticks* in a population and becomes the “new normal”.

Substitution models describe the *probability* of these changes happening along the branches of a phylogenetic tree.



Rate Matrix

Every substitution model is defined by a **rate matrix (Q)**, which tells us how fast different changes happen

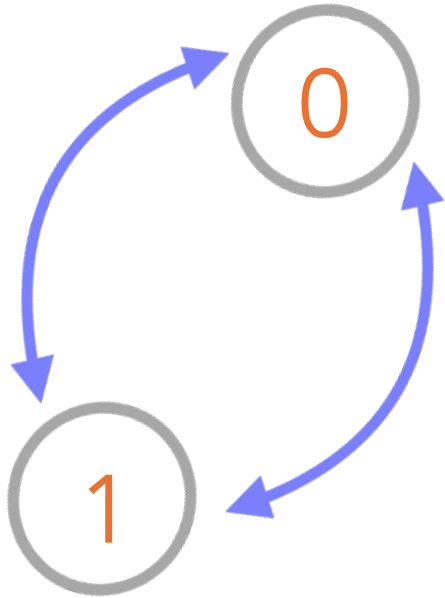
For DNA: 4 possible states (A, C, G, T).

For proteins: 20 states (amino acids).

For morphology: as many states as you code for a character.

Different substitution models are defined using the Q matrix.

Rate Matrix



$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{pmatrix}$$

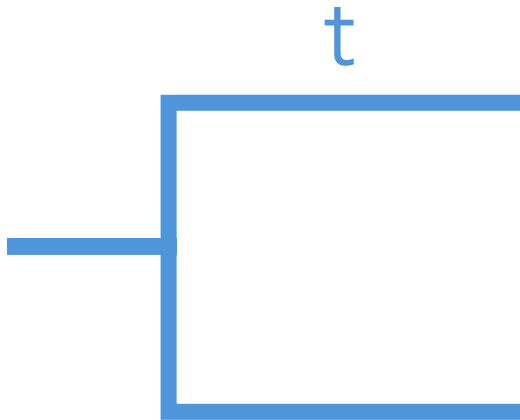
Must sum to zero

Any assumptions about your data
can be incorporated through the
mathematical expression

Rate Matrix

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{pmatrix}$$

This tells us the rate - how can we use this during inference?



Matrix
exponentiation

$$P(t) = e^{Qt}$$

This allows us to calculate the probability of a switch along a branch

Bayes Theorem

The **rate matrix (Q)** is what gives us the **likelihood term**.
The data at the tips (DNA bases, amino acids, or morphology) have to be explained by the tree.

$$P(\text{Tree} \mid \text{Data}) = \frac{P(\text{Data} \mid \text{Tree}) P(\text{Tree})}{P(\text{Data})}$$

The diagram illustrates Bayes' Theorem for phylogenetic inference. The left side represents the joint probability of a tree and data. The right side is the product of the likelihood and the prior probability of the tree, divided by the marginal probability of the data.

Diagram Details:

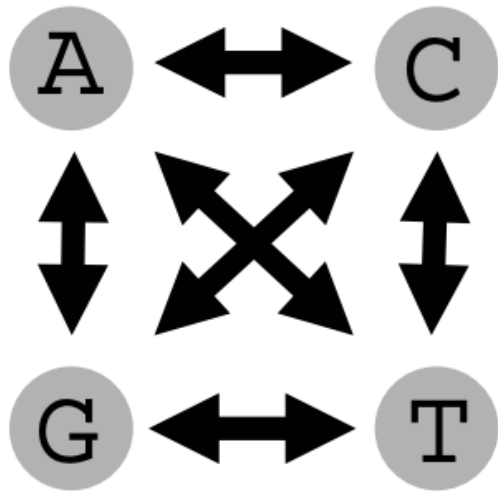
- Tree:** A simple three-taxon tree with a root and three tips.
- Rate Matrix (Q):** A 2x2 matrix with states 0 and 1. The diagonal elements are 0. The off-diagonal elements are labeled with arrows: 0 to 1 (top-right) and 1 to 0 (bottom-left).
- Data:** A 3x1 vector of binary characters (0 or 1) at the tips of the tree.

Tree	Q	Data
Tree 1	Q	0101...
Tree 2	Q	1101...
Tree 3	Q	0100...

How probable it is to see your fossil and/or molecular data for a given tree?

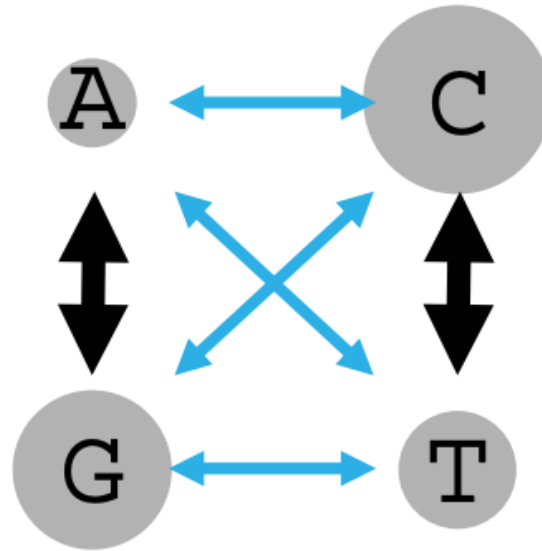
Substitution models in molecular data

JC



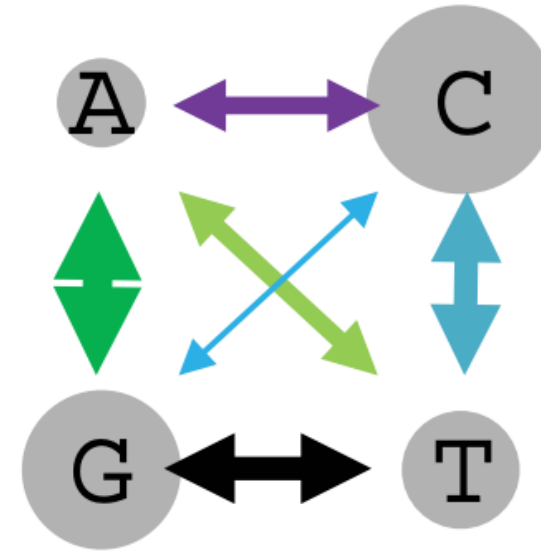
All changes equal
Equal base frequencies

HKY



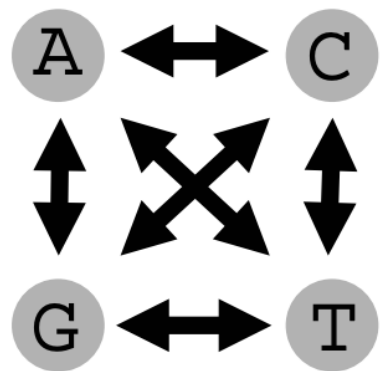
Transitions $\neq$ Transversions
Unequal base frequencies

GTR



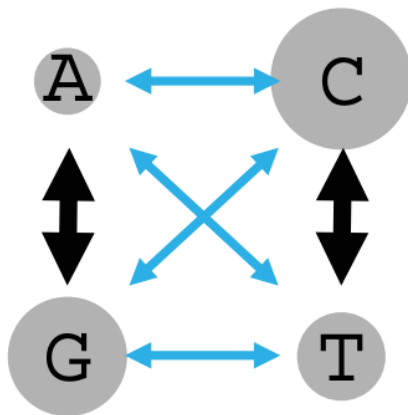
Every substitution type has its own rate
Unequal base frequencies

JC



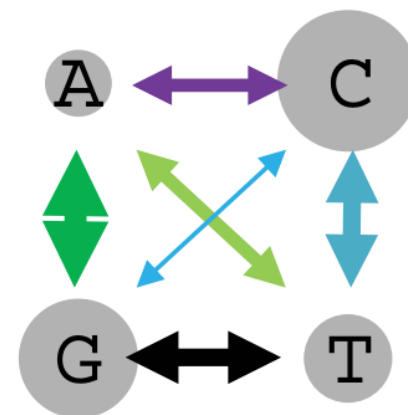
$$Q = \begin{pmatrix} * & \frac{\mu}{4} & \frac{\mu}{4} & \frac{\mu}{4} \\ \frac{\mu}{4} & * & \frac{\mu}{4} & \frac{\mu}{4} \\ \frac{\mu}{4} & \frac{\mu}{4} & * & \frac{\mu}{4} \\ \frac{\mu}{4} & \frac{\mu}{4} & \frac{\mu}{4} & * \end{pmatrix}$$

HKY



$$Q = \begin{pmatrix} * & \kappa\pi_G & \pi_C & \pi_T \\ \kappa\pi_A & * & \pi_C & \pi_T \\ \pi_A & \pi_G & * & \kappa\pi_T \\ \pi_A & \pi_G & \kappa\pi_C & * \end{pmatrix}$$

GTR

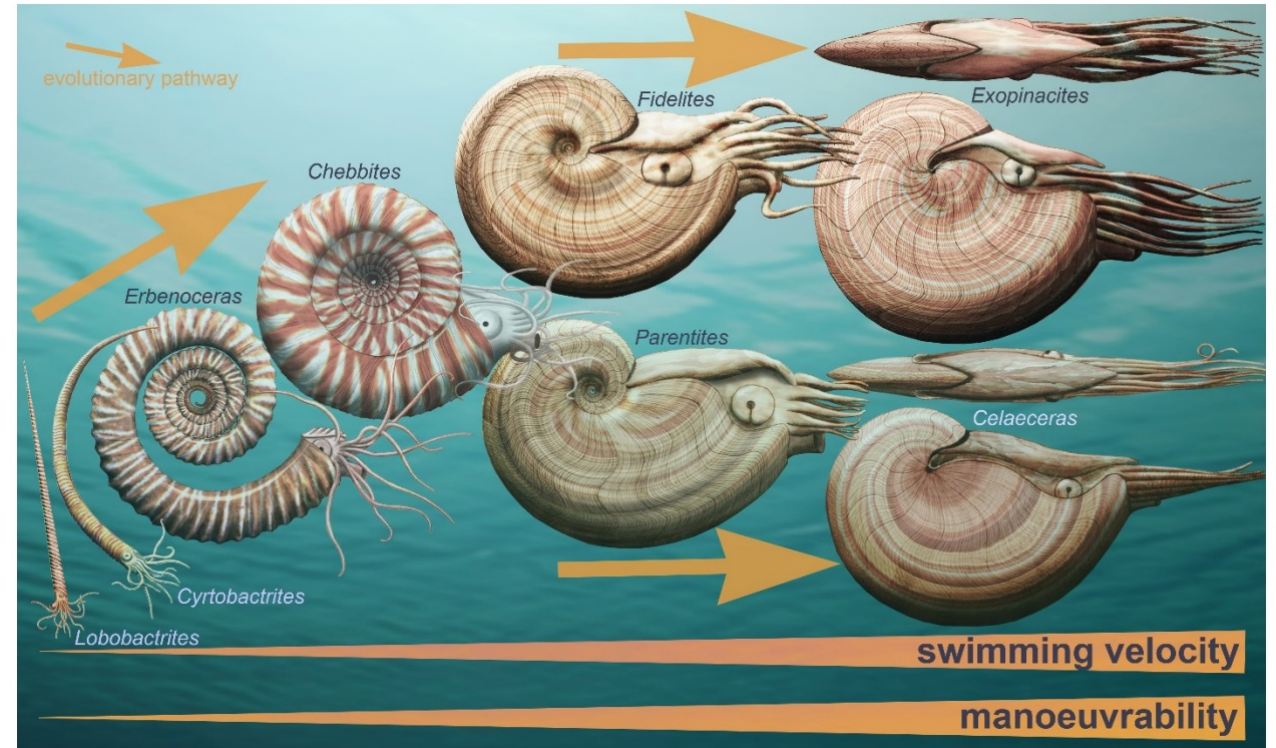


$$Q = \begin{pmatrix} * & \kappa_1\pi_G & \pi_C & \pi_T \\ \kappa_1\pi_A & * & \pi_C & \pi_T \\ \pi_A & \pi_G & * & \kappa_2\pi_T \\ \pi_A & \pi_G & \kappa_2\pi_C & * \end{pmatrix}$$

Morphological data

Morphological data was the original type of information used in phylogenetic analysis.

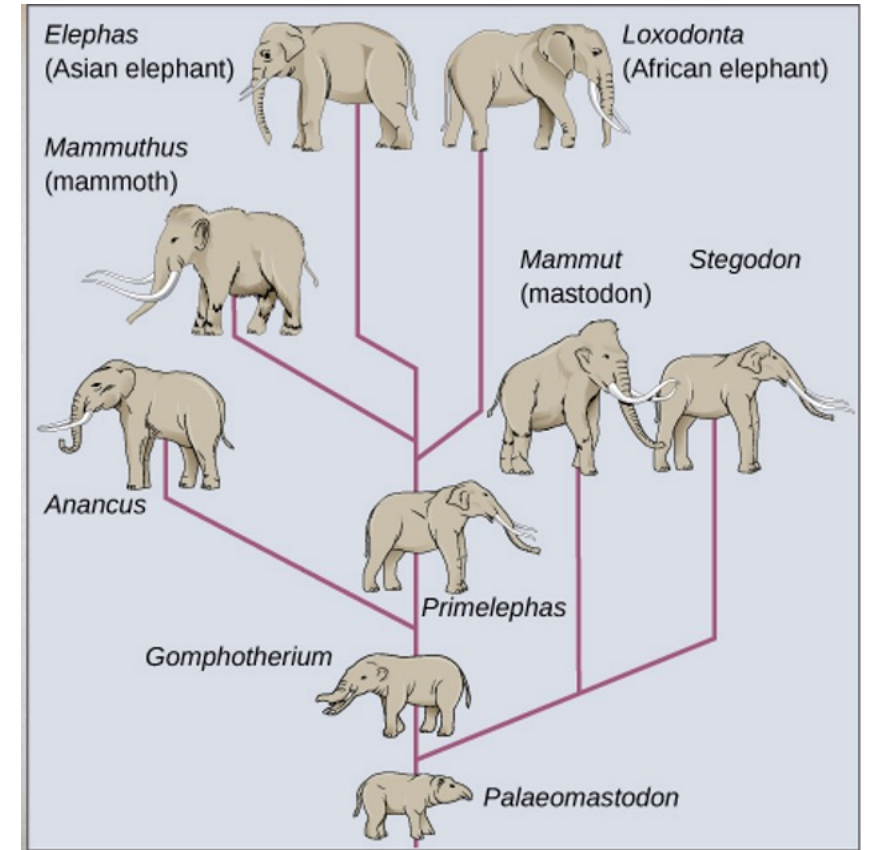
Fossils can be used to provide time calibrations, helps refine extant phylogenies, allows us to understand evolution through time.



Types of morphological data

Discrete Characters: Morphological data often consist of discrete characters, such as the presence or absence of certain traits, or more complex multistate traits (e.g., number of limbs, type of leaf, presence of a particular bone structure)

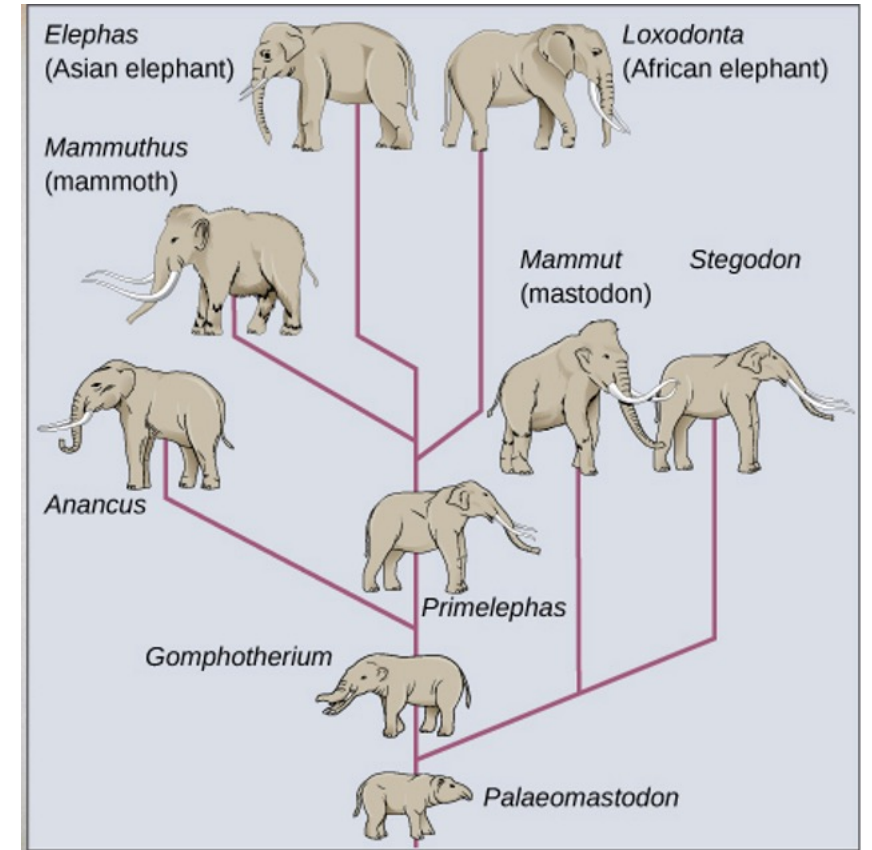
Continuous Characters: Some morphological data can be continuous, such as measurements of body size, length of bones, or other quantitative traits



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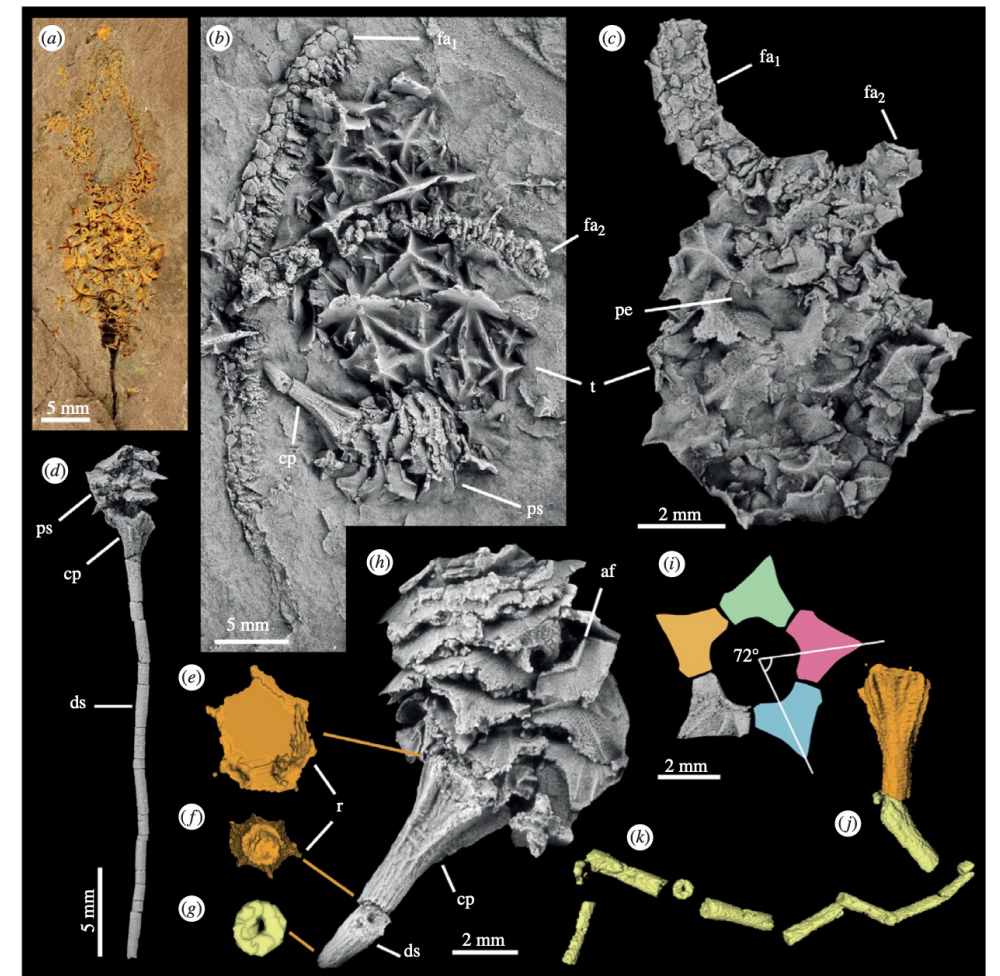
Continuous Characters: Some morphological data can be continuous, such as measurements of body size, length of bones, or other quantitative traits



Trait 1

Trait 28

Taxa 1	001510010?00-100--00000000000
	000500010?200100--00100100000
	002500010?200100--0?100100000
	00?5?0010?200100?-0???010110
	0015000101201000430100011111
	0015000101201010440111011111
	??050?????201000440?11011111
	01050?010-210000?501??010110
	00020001002101003-1110010110
	0002000100211001441121011111
	000201111-210010?-??11011121
	?103?0?11?1001104-0000010000
	1005002110100010--0?00110?20
Taxa 14	1005002000101010540?00110020



Cambrian stalked echinoderms show unexpected plasticity of arm construction
Zamora & Smith. 2012 Proc B

Making
general
assumptions
about
evolution
from this
data is
extremely
difficult

Multistate
characters

Non-applicable
characters

Presence/
absence

```
101510010?00-100--000000000000
000500010?200100--00100100000
102500010?200100--0?100100000
00?5?0010?200100?-0??010110
0015000101201000430100011111
0015000101201010440111011111
??050????201000440?11011111
```

Often small

Discrete character data

[illegible]

Discrete character data

Binary traits	0 1	Often describes the presence/absence of a trait
Multistate traits	0 1 2 3 4	Used to describe more complex traits and can capture greater variation between taxa

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Polymorphisms	0/1/2	Used when there are variations in a traits within species
Uncertain	0/1/2	Used when it is not clear which character trait is present in the taxon

How do we model
morphological
evolution?

Mk model

Assumes **equal transition probabilities** between states

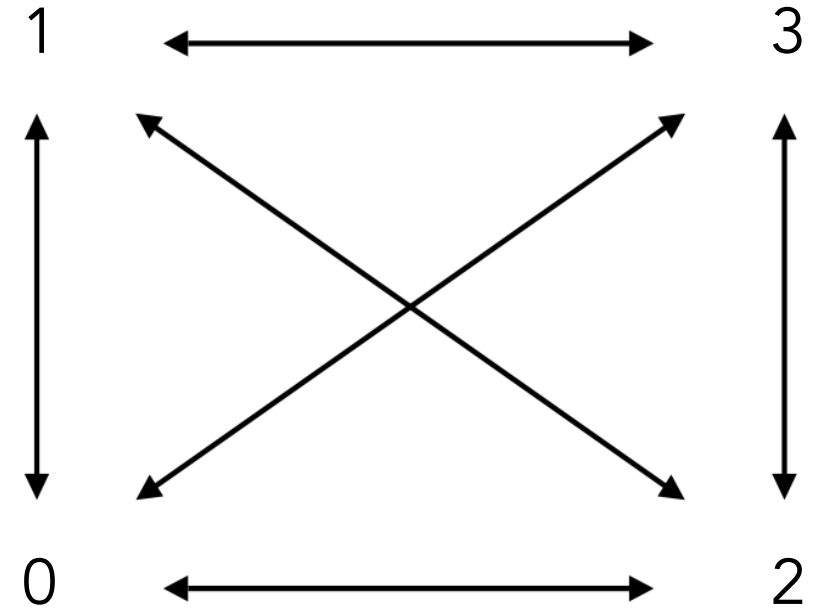


$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{pmatrix}$$

Mk model

K can be any number of states

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$



*4 state here as an example, can be any number from 2!

MkV model

What is one
characteristic of
morphological data
that is extremely
different to molecular?

```
001510010?00-100--00000000000
000500010?200100--0010010000
002500010?200100--0?10010000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050?????201000440?11011111
01050?010-210000?501??010110
00020001002101003-1110010110
0002000100211001441121011111
000201111-210010?-??11011121
?103?0?11?1001104-0000010000
1005002110100010--0?00110?20
1005002000101010540?00110020
```

MkV model

What is one
characteristic of
morphological data
that is extremely
different to molecular?

All varying characters

```
001510010?00-100--00000000000
000500010?200100--00100100000
002500010?200100--0?100100000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050?????201000440?11011111
01050?010-210000?501??010110
00020001002101003-1110010110
0002000100211001441121011111
000201111-210010?-??11011121
?103?0?11?1001104-0000010000
1005002110100010--0?00110?20
1005002000101010540?00110020
```

MkV model

Corrects for **ascertainment bias**

Failing to account for this can lead to **overestimation of branch lengths** and which can further lead to errors in topology!



Condition the likelihood
on there only being
varying site

$$\Pr (D \mid V) = \frac{\Pr (D, V)}{\Pr (V)}$$

MkV model

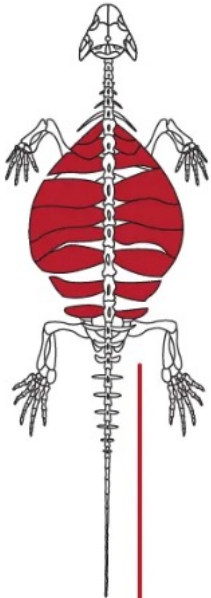


	True Branch Length	Mk	Mkv
Percent correct	-	74.0	99.8
Branch A	0.2	241,750 (±349,100)	0.206 (±0.060)
Branch B	0.05	0.43210 (±0.13756)	0.050 (± 0.018)
Branch X	0.05	54.646 (±1,725.3)	0.052 (± 0.023)
Branch C	0.2	143,950 (±228,910)	0.206 (± 0.059)
Branch D	0.05	0.022 (± 0.054)	0.051 (±0.019)

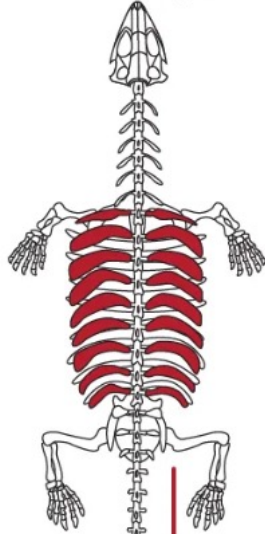
Among character rate variation

Turtle shell evolution

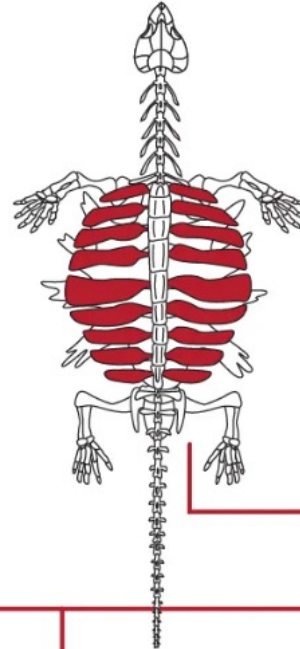
Eunotosaurus
~260 mya



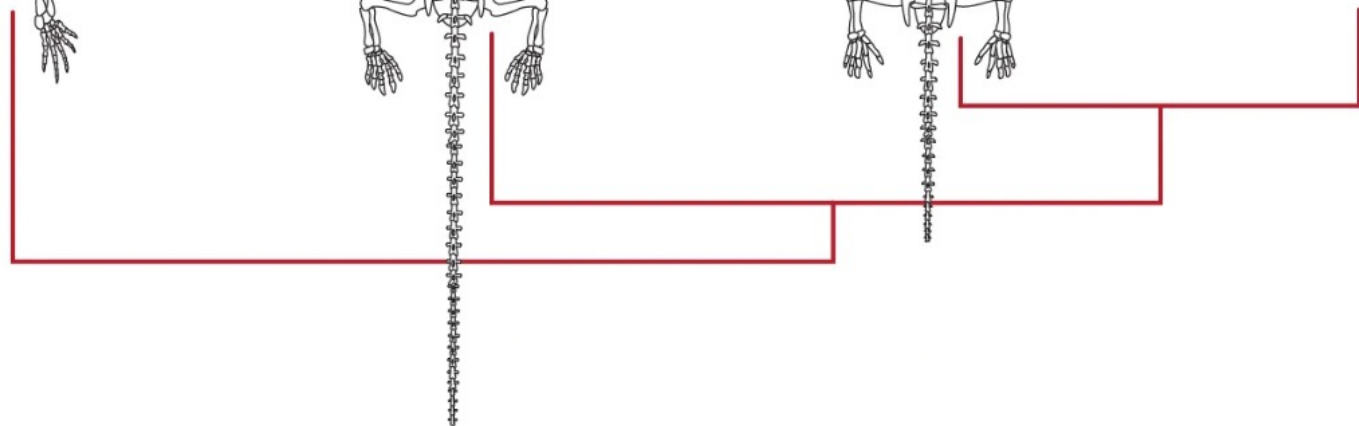
Pappochelys
~240 mya



Odontochelys
~220 mya



Proganochelys
~210 mya



Among character rate variation

	Trait 1	Trait 2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

The transition rate
will impact branch
lengths

Slow rate of evolution



Fast rate of evolution

Relative to each other!

Among character rate variation

What do we do?

	T1	T2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

Allow these traits to evolve at different rates:

- Specify which traits evolve fast
- Use a gamma model to account for rate heterogeneity

Among character rate variation

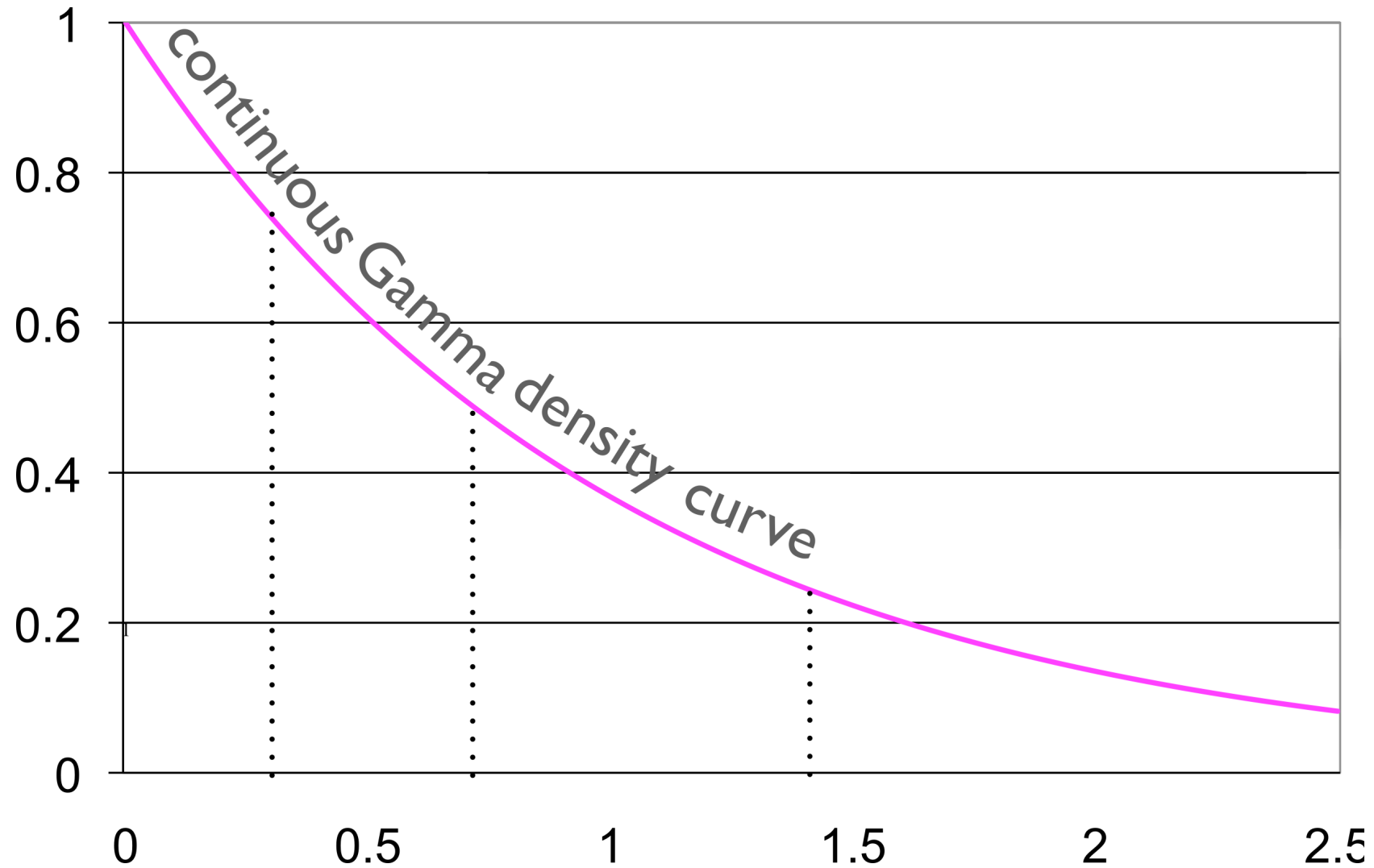
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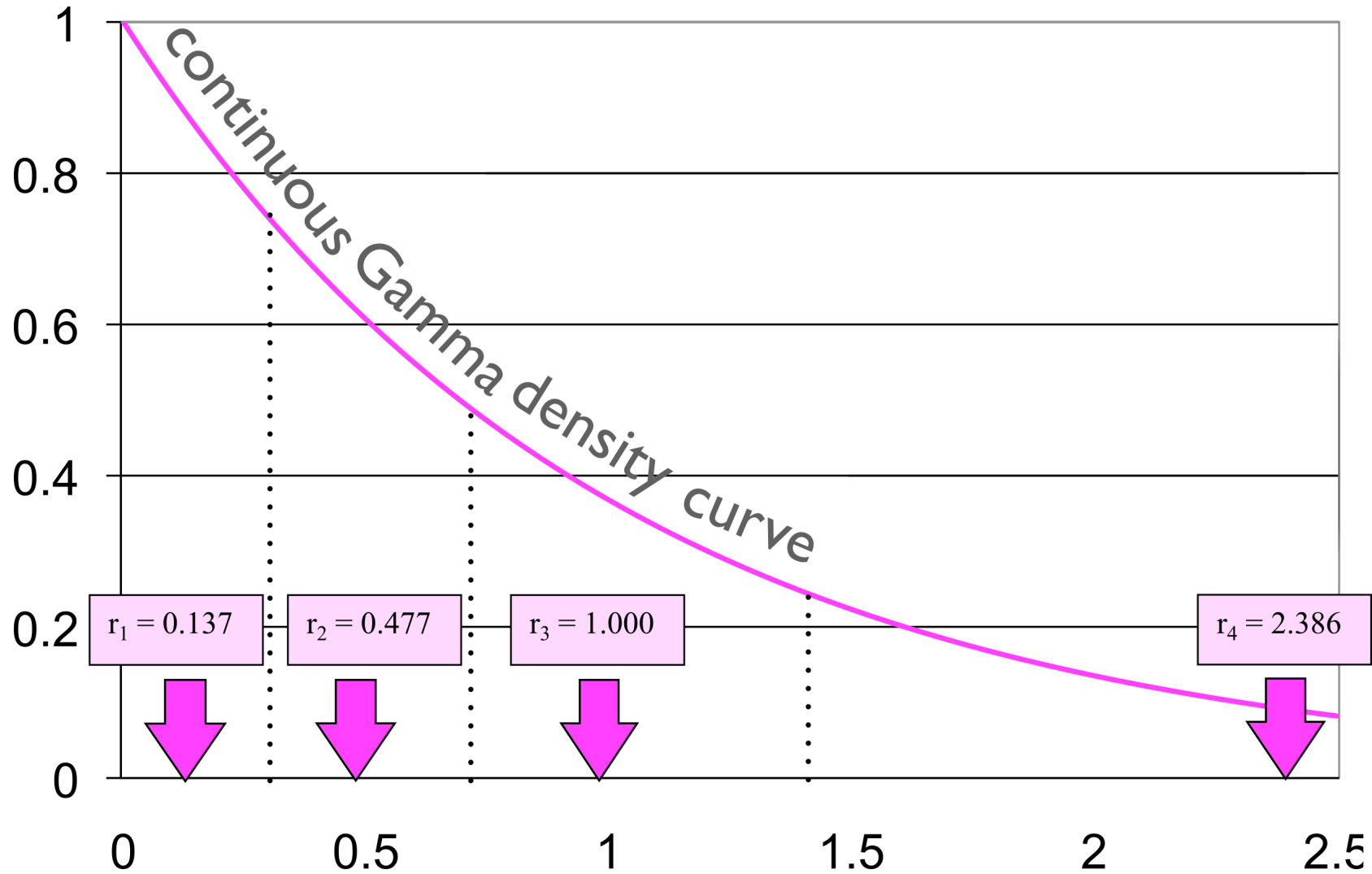
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Among character rate variation



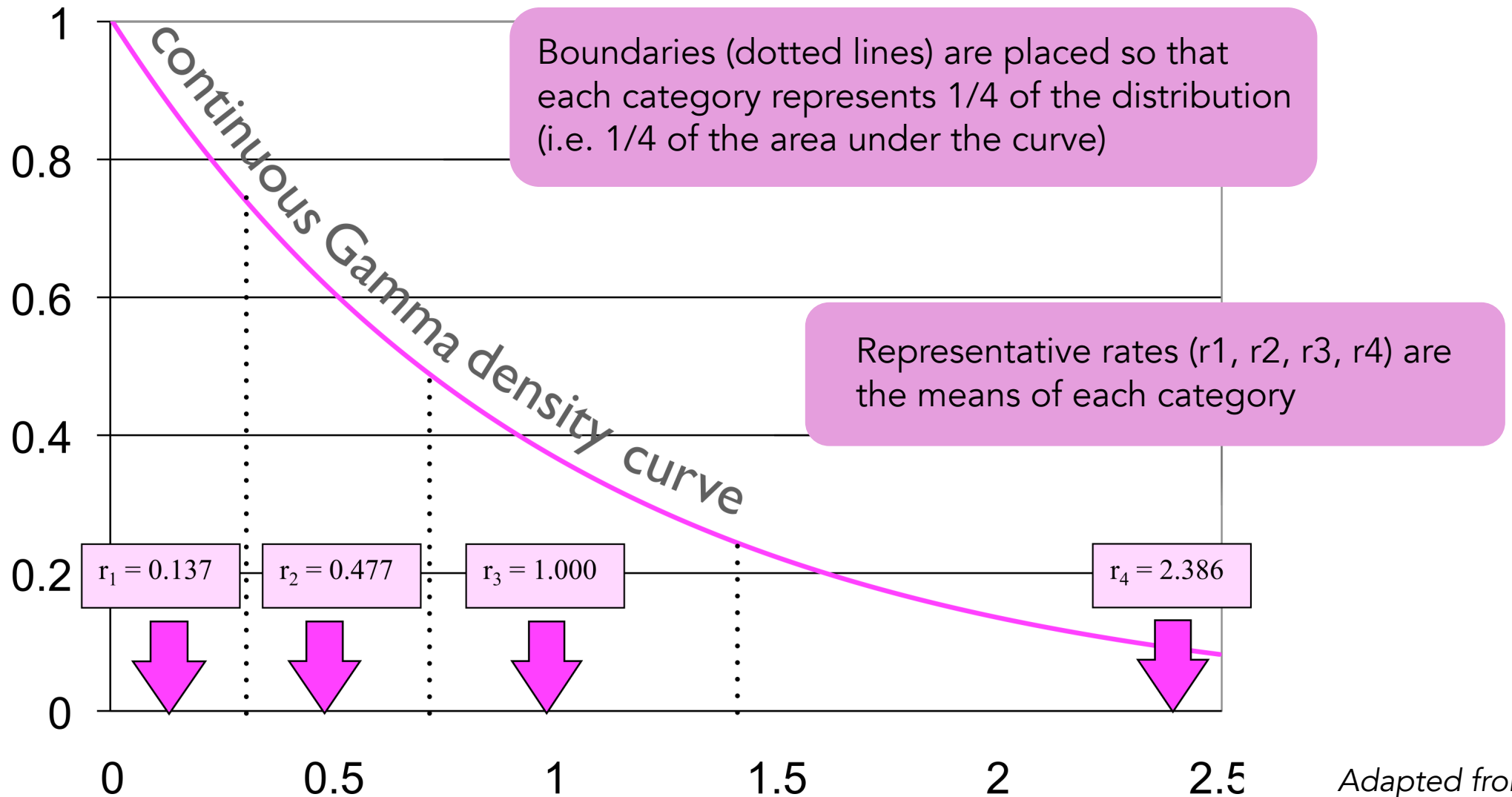
Adapted from Paul
Lewis PhyloSeminar

Among character rate variation



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Among character rate variation



Among character rate variation

What do we do?

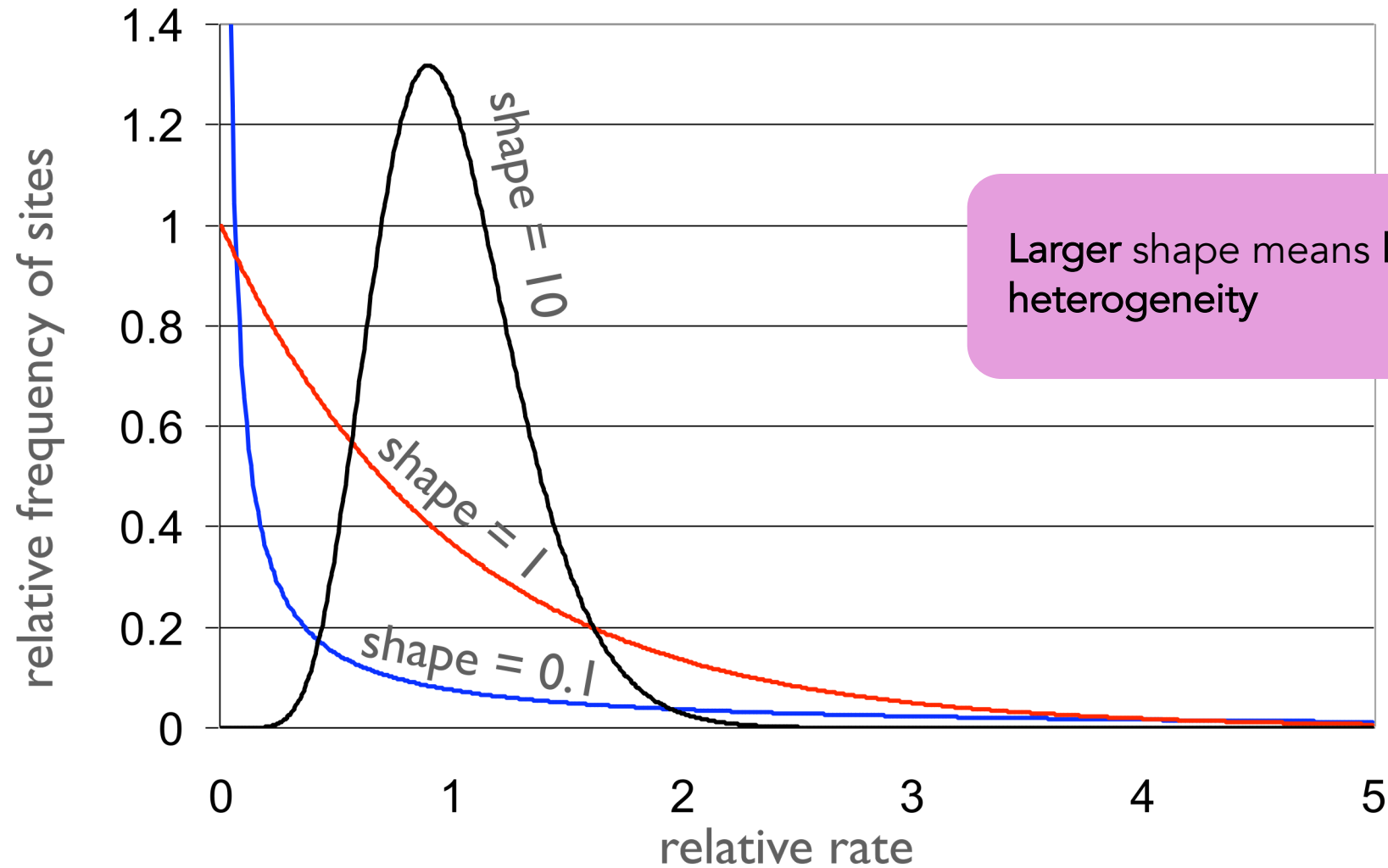
	T1	T2	
Taxa A	0	0	
Taxa B	0	1	
Taxa C	1	2	Faster R (R4)

Slower R (R1,2)

Allow each trait to evolve according to the rates drawn from the gamma distribution.

One rate will fit the best and be the most influential for the likelihood calculation.

Among character rate variation



Incorporating ACRV into the matrix

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix} * r$$

Compute the likelihood for each site

$$L_1 = P(\text{site} \mid Q_1)$$

$$L_2 = P(\text{site} \mid Q_2)$$

$$L_3 = P(\text{site} \mid Q_3)$$

$$L_4 = P(\text{site} \mid Q_4)$$

For each category:

$$Q_1 = r_1 \cdot Q,$$

$$Q_2 = r_2 \cdot Q,$$

$$Q_3 = r_3 \cdot Q,$$

$$Q_4 = r_4 \cdot Q$$

Average over categories

$$L_{\text{site}} = 1/4 (L_1 + L_2 + L_3 + L_4)$$

This gives the **final likelihood for that site**, accounting for among-site rate variation.

Partitioning data

Grouping together parts of the matrix that have similar characteristics and/or may have **evolved together** due to evolutionary pressures.

The **defaults** in many phylogenetic software is to group by maximum observed state size.

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{pmatrix}$$

Partitioning data

When should we partition our data?

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If we have presence (1) absence (0) traits partitioning will always be a logical approach: what would transitioning to state 2 in this scenario even mean?

Partitioning data

When should we partition our data?

If we have presence (1) absence (0) traits partitioning will always be a logical approach: what would transitioning to state 2 in this scenario even mean?

We should be cautious for states describing a trait – just because we do not observe a state 2 can we be absolutely certain there never was one?

Justifying partitioning schemes is very important as they have a major impact on inference results

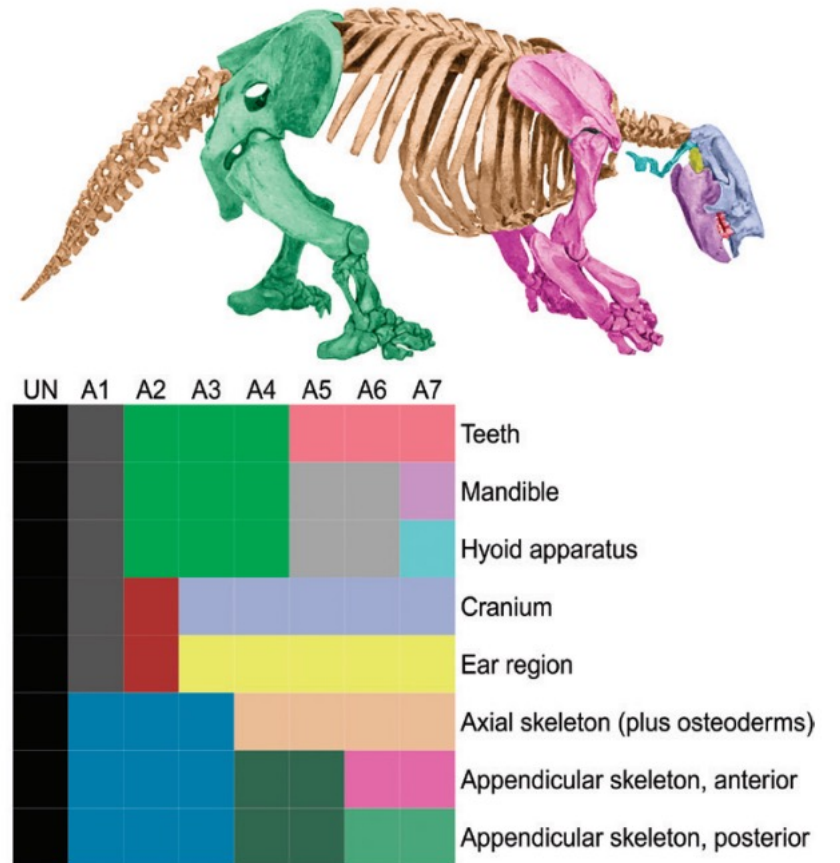
Other morphological
models

Alternative partitioning schemes

Reassessing the phylogeny and divergence times of sloths (Mammalia: Pilosa: Folivora)

Characters can be groups based on anatomical region

Other criteria such as the degree of homoplasy present in a character was explored in this study – and found to be a better fit using Bayes factors



Ordered characters

Ordered characters can be placed in an order so that transitions only occur between adjacent states.



For example, “**intermediate**” species that are somewhere in between limbed and limbless – for example, the “mermaid skinks” (*Sirenoscincus*) from Madagascar, so called because they lack hind limbs. An ordered model might only allow transitions between limbless and intermediate, and intermediate and limbed; it would be impossible under such a model to go directly from limbed to limbless without first becoming intermediate.

For unordered characters, any state can change into any other state.

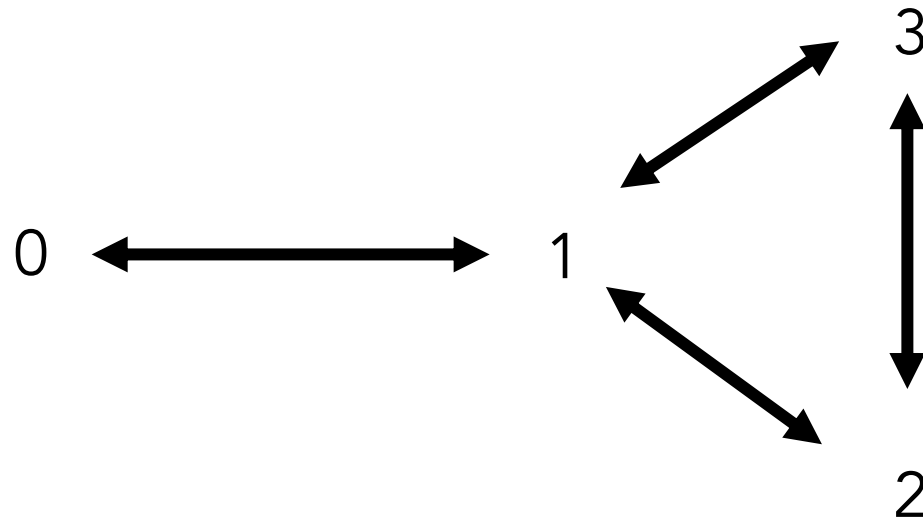
Ordered characters



All characters ordered:



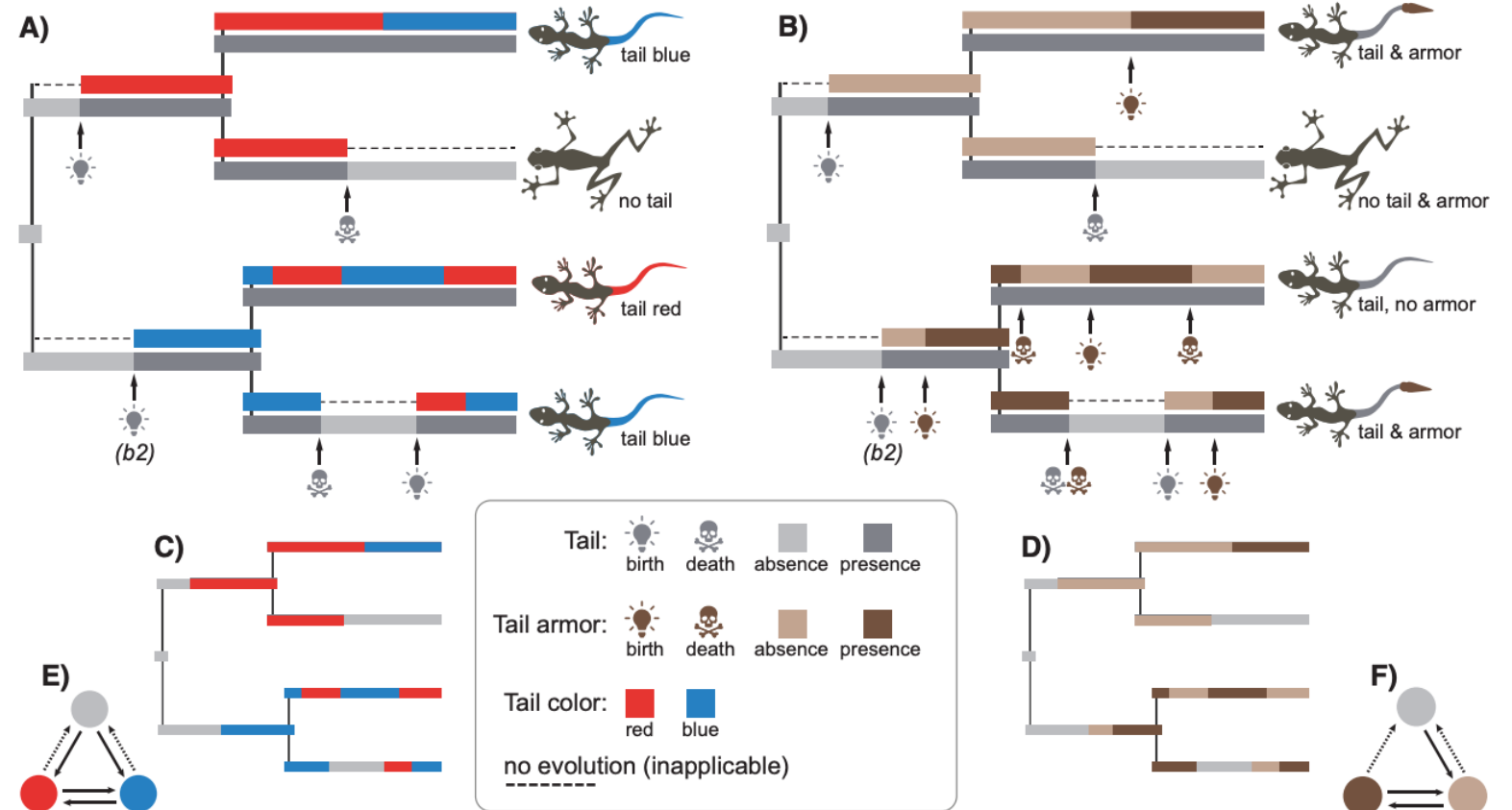
Specific characters
ordered:



Embedded dependency model

Markov models for
phylogenetic inference
with anatomically
dependent (inapplicable)
morphological characters

Non-applicable characters
only considered when
they are present (1)



Challenges with morphological data

Generalising assumptions across different traits is often not possible.

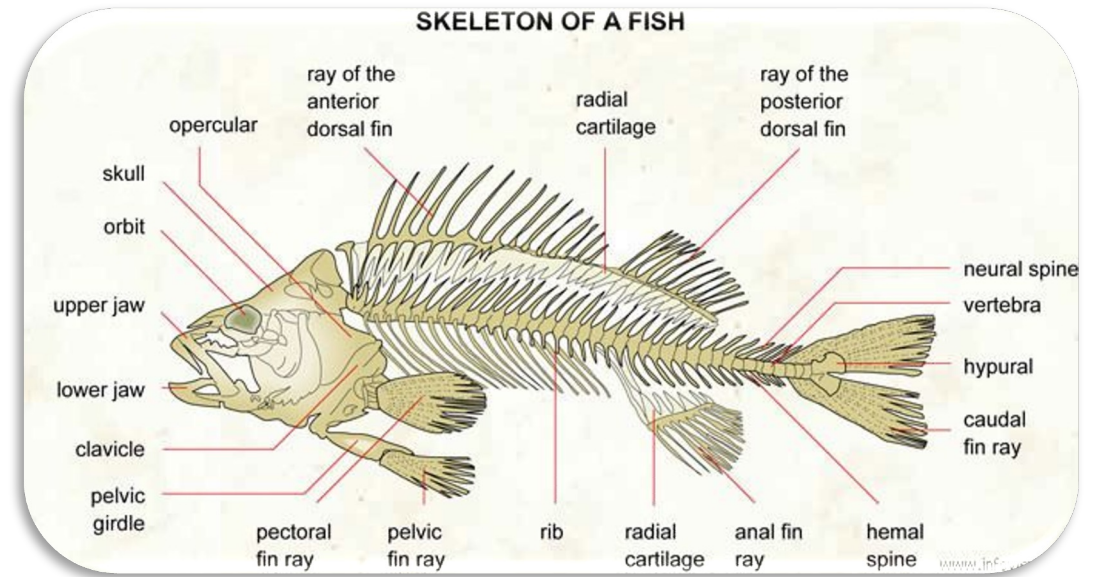
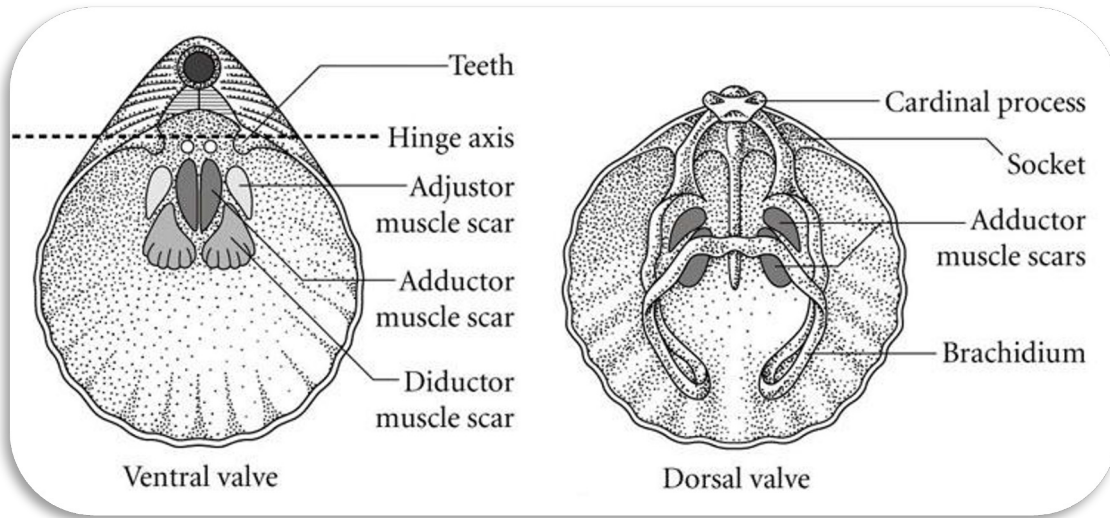
Character correlation occurs when two or more characters are **not independent**. Functional/developmental linkage: Traits are biologically linked. Example: The length of finger bones may be correlated with the length of the hand.

```
101510010?00-100--00000000000
000500010?200100--0010010000
102500010?200100--0?10010000
00?5?0010?200100?-0???010110
0015000101201000430100011111
```

Challenges with morphological data

Morphological matrices are often quite small:

- Collection is very time consuming
- Number of characters available can be very small depending on the group

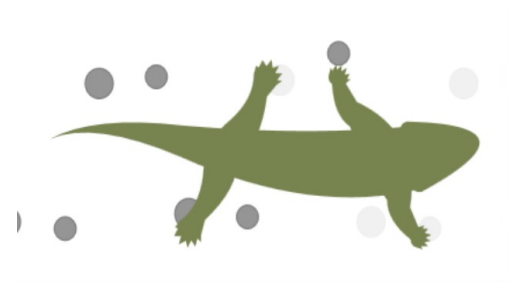


Impact of model on key parameter estimates

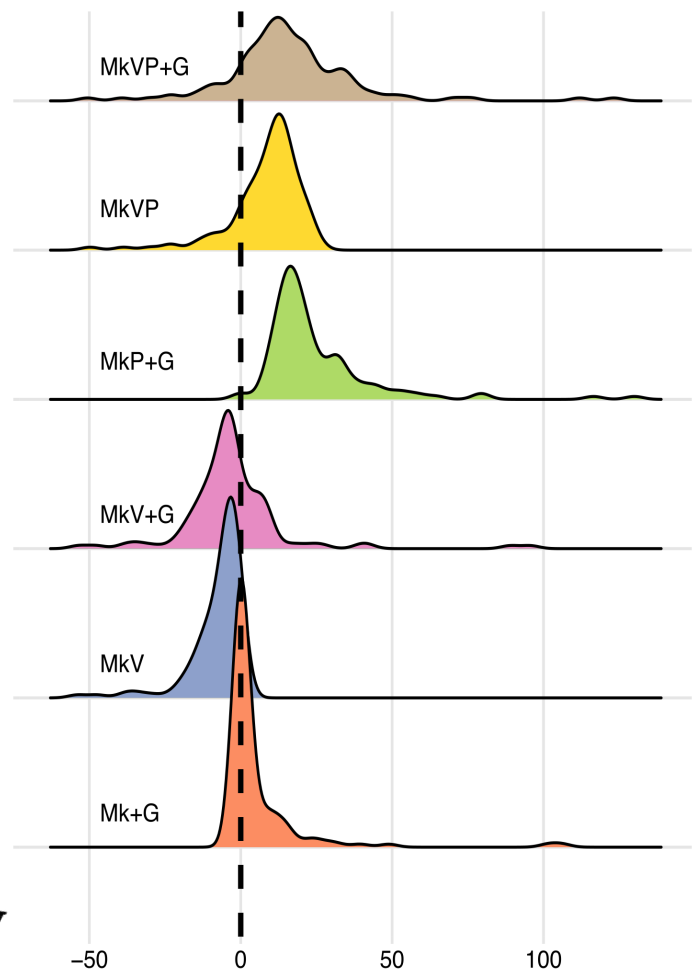
Example of 114 empirical **tetrapod** matrices

Looked at the impact on:

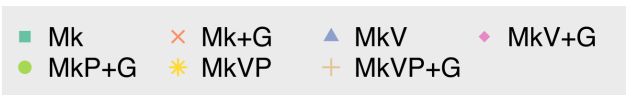
- branch lengths (**evolutionary distances**)
- Tree topology (**species relationships**)

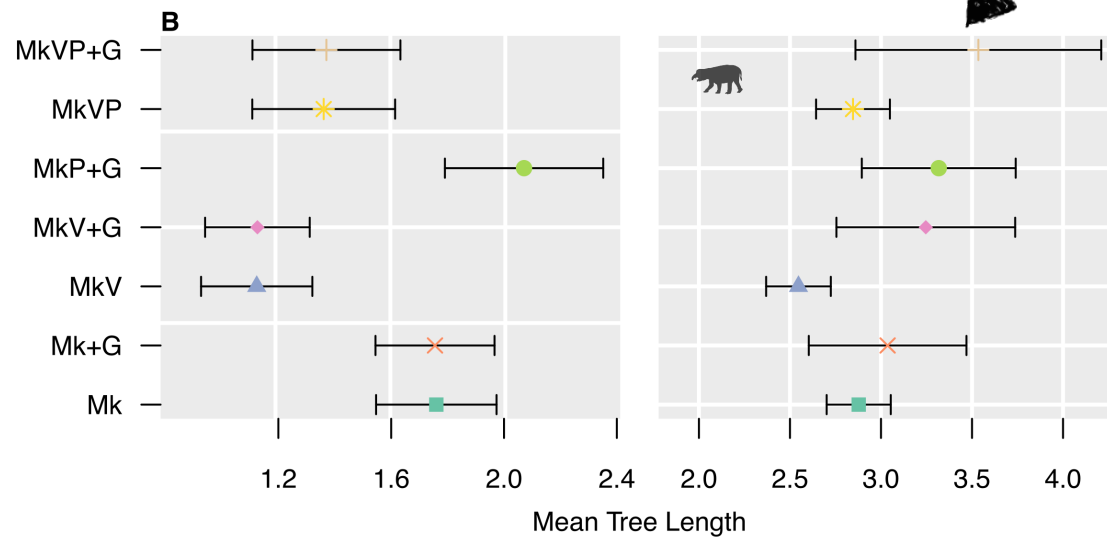
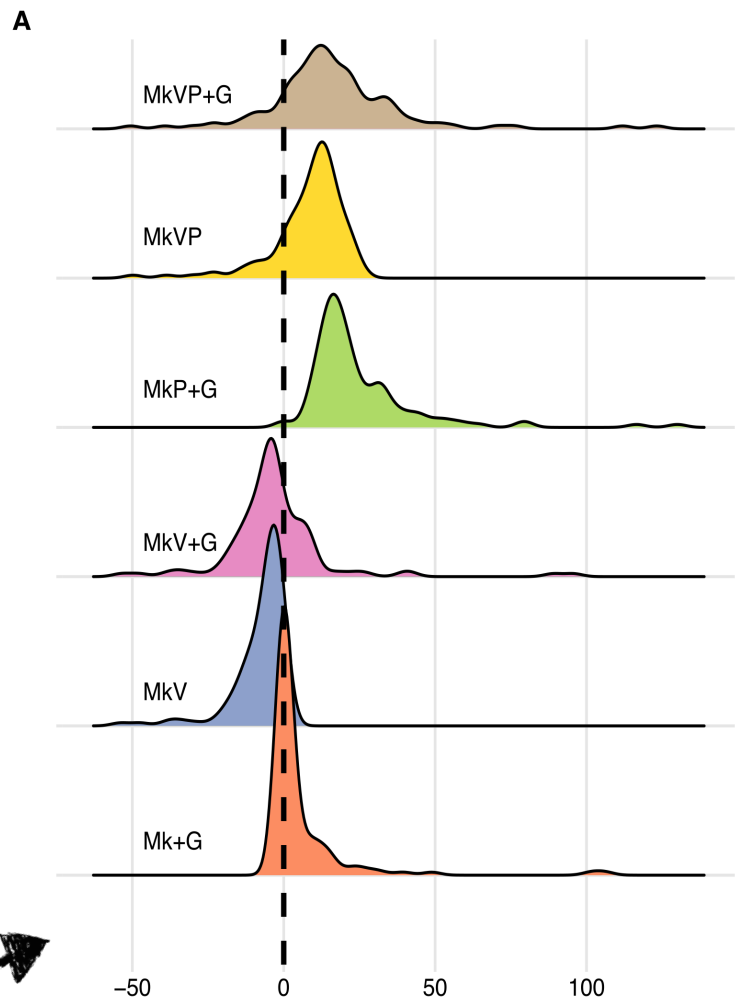


A

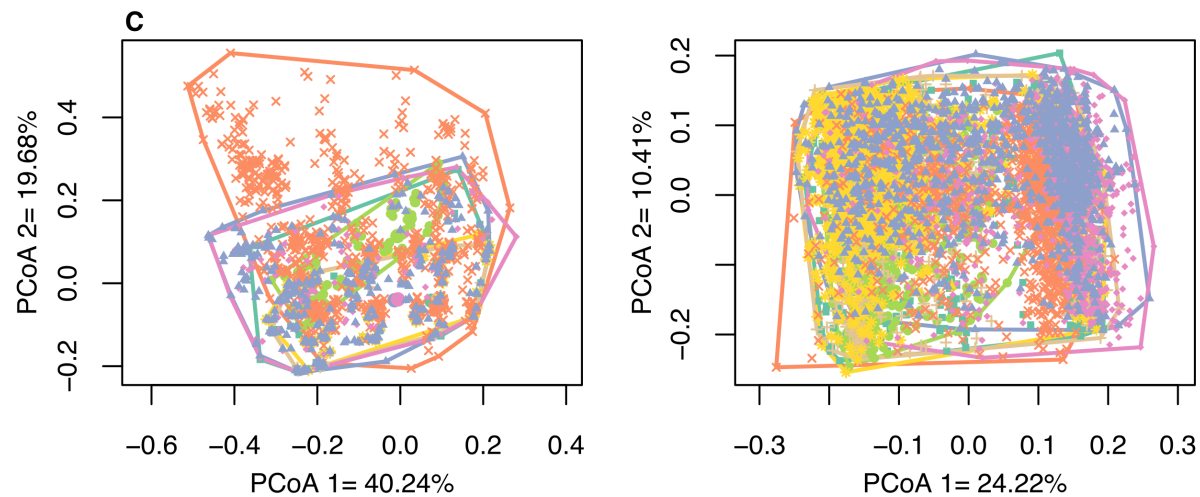


Percentage difference in
tree length relative to Mk
model





Tree length of
two different
data sets



Rf
distances
of two
data sets

Percentage difference in
tree length relative to Mk
model



How do we choose a
model?

Model selection

Bayes factors are commonly used to determine the **relative fit** between models.

It relies on comparing the marginal likelihoods approximated from different models.

The ML measures the average fit of a model to our data.

We use MCMC to avoid calculating this number as it is computationally expensive and often not directly possible.

Model selection

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model}) P(\text{model})}{P(\text{data} \mid \text{model})}$$

Diagram illustrating the components of the model selection equation:

- Posterior**: Points to $P(\text{model} \mid \text{data})$
- Likelihood**: Points to $P(\text{data} \mid \text{model})$ in the numerator
- Priors**: Points to $P(\text{model})$ in the numerator
- Marginal likelihood**: Points to $P(\text{data} \mid \text{model})$ in the denominator

Marginal likelihood

Marginal probability of the data (denominator in Bayes' rule) is the expected value of the likelihood with respect to the prior distribution.

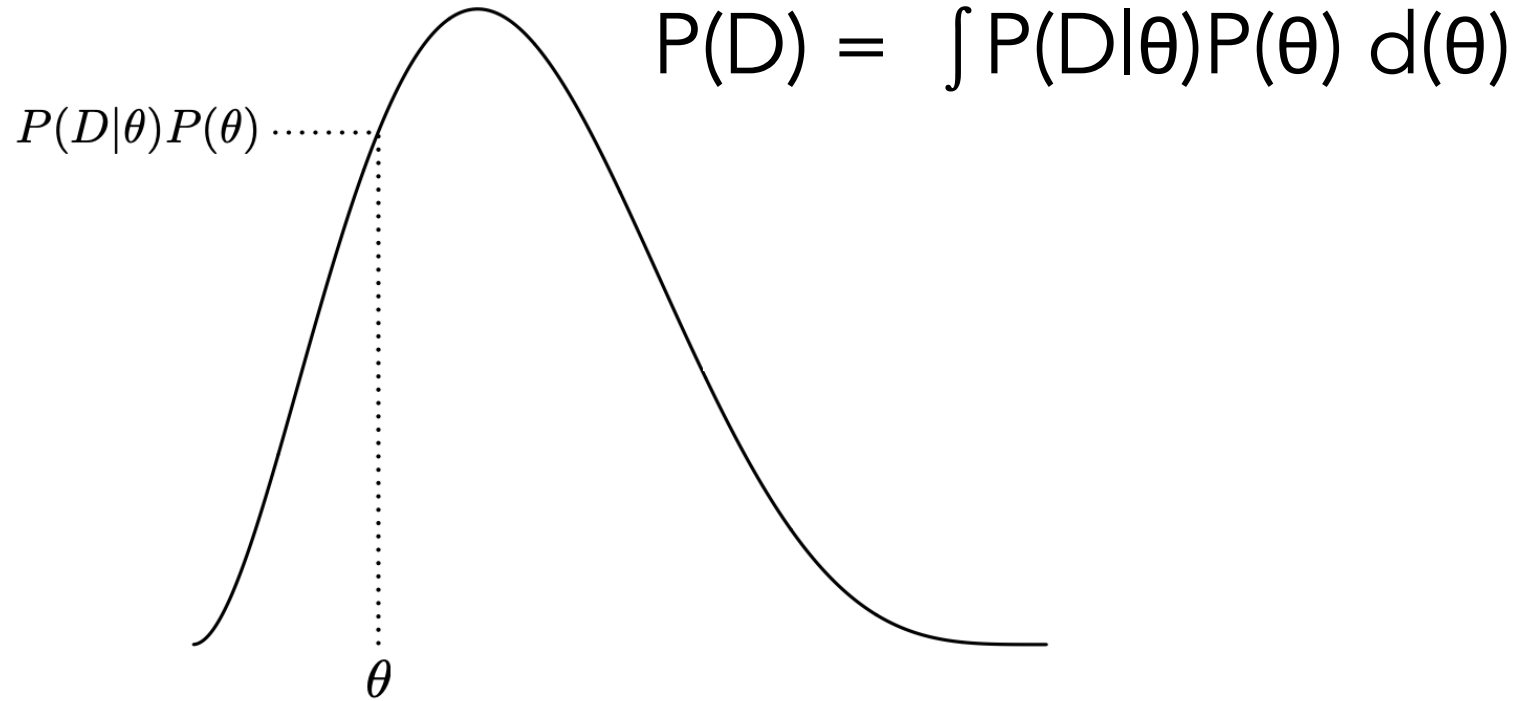
If likelihood measures model fit, then the marginal likelihood measures the **average fit** of the model to the data over all parameter values.

What is the expected value?

Marginal likelihood

$P(\text{data} \mid \text{model})$

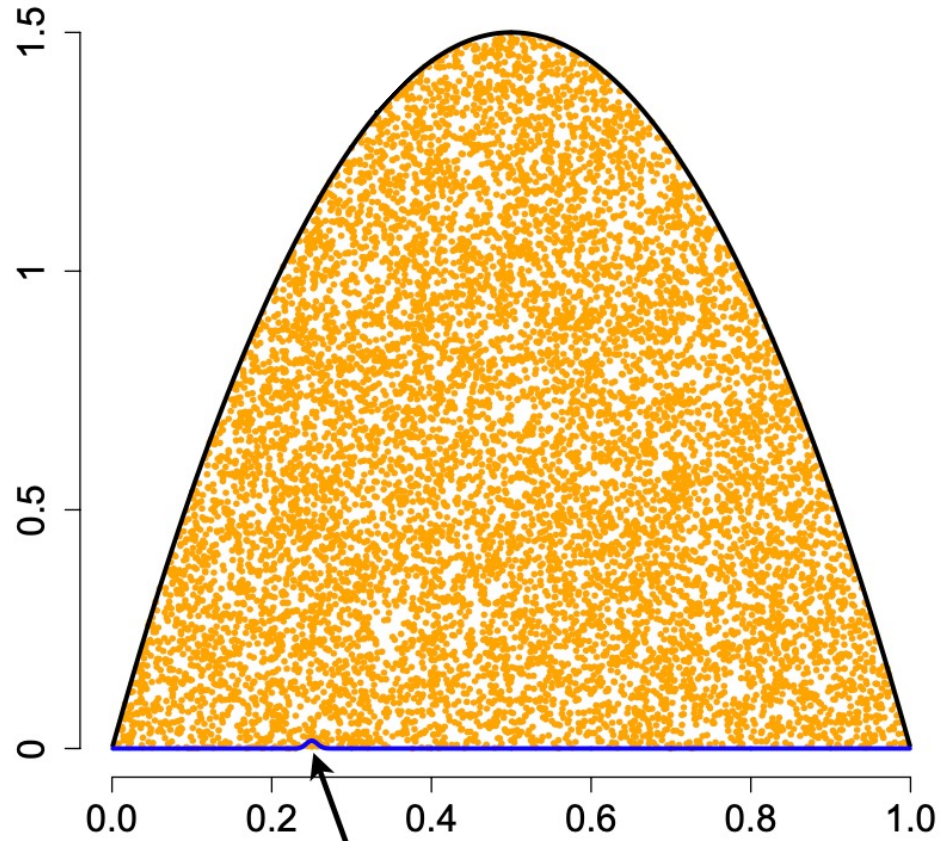
The marginal likelihood is used to evaluate the overall fit of the model to the data, integrating over all parameter values.



Marginal likelihood

$P(\text{data} \mid \text{model})$

Very small, single
number between the
posterior distribution
and the prior



Approximating the marginal likelihood

There are two common algorithms to do this:

- Stepping stone
- Path sampling

Both of these approaches are computationally expensive.

Stepping-stone algorithms are like a series of MCMC simulations that iteratively sample from a specified number of distributions that are discrete steps between the posterior and the prior probability distributions.

Bayes factors

$$B_{01} = \frac{P(D | M_0)}{P(D | M_1)} = \frac{\text{Marginal likelihood for model } M_0}{\text{Marginal likelihood for model } M_1}$$

Bayes factors

$$B_{01} = \frac{P(D | M_0)}{P(D | M_1)} = \frac{\text{Marginal likelihood for model } M_0}{\text{Marginal likelihood for model } M_1}$$

Marginal likelihoods are often on the log scale so the Bayes factor can be calculated as:

$$\log B_{01} = \log P(D | M_0) - \log P(D | M_1)$$

Bayes factors

Strength of evidence	$BF(M0, M1)$	$\log(BF(M0, M1))$
Negative (supports M_1)	<1	<0
Barely worth mentioning	1 to 3.2	0 to 1.16
Substantial	3.2 to 10	1.16 to 2.3
Strong	10 to 100	2.3 to 4.6
Decisive	>100	>4.6

Issues with Bayes factors for morphological data

The way we **partition data** for morphological data is different to molecular

0	1	0	0	2	3
2	0	1	1	0	2
1	1	2	1	3	1

Unpartitioned everything
in Q-matrix of size 4

Partitioning the data puts
characters into correctly
sized Q-matrix

The way we **partition data** for morphological data is different to molecular

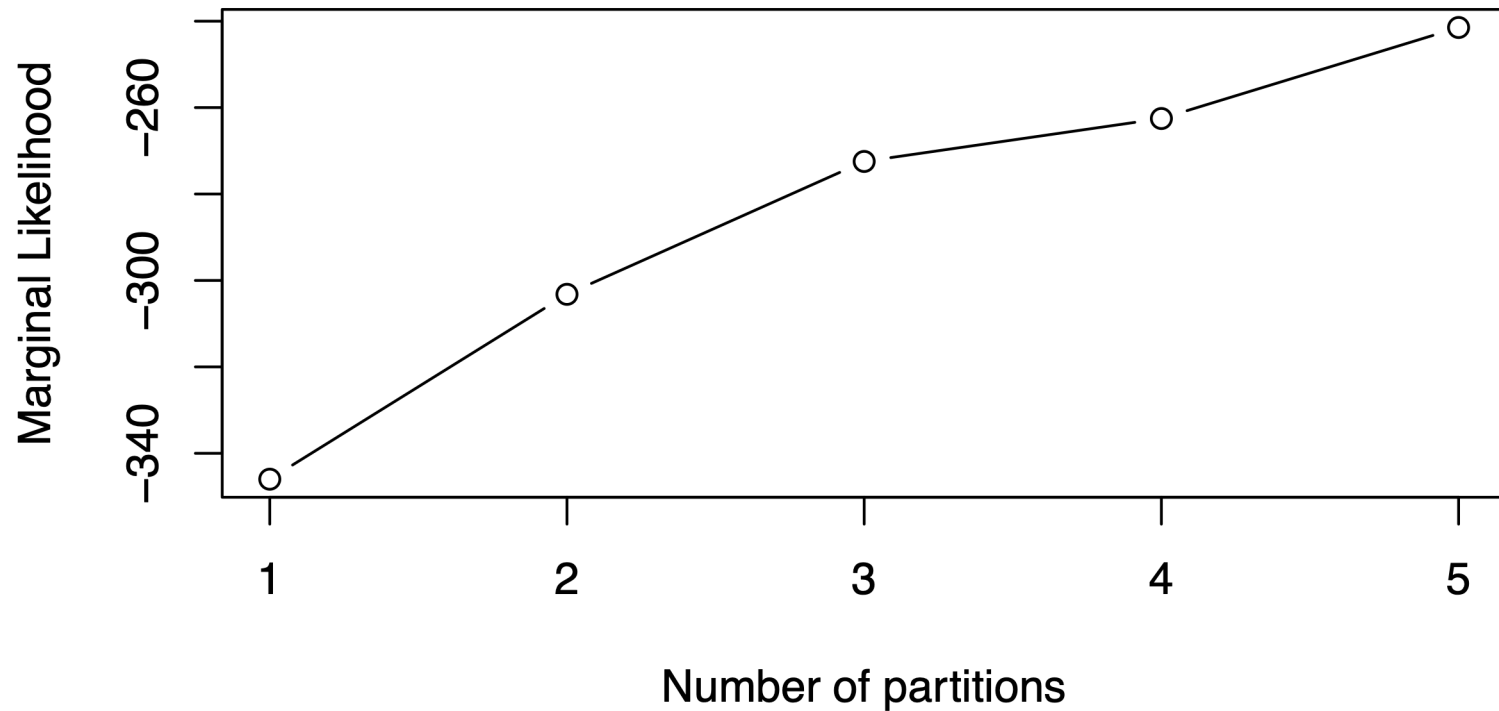
010023
201102
112131

10	00	23
01	21	02
11	12	31

Unpartitioned everything in Q-matrix of size 4

Partitioning the data puts characters into correctly sized Q-matrix

Issues with Bayes factors for morphological data



Data set with 6 states

As the number of partitions increases, so does the likelihood

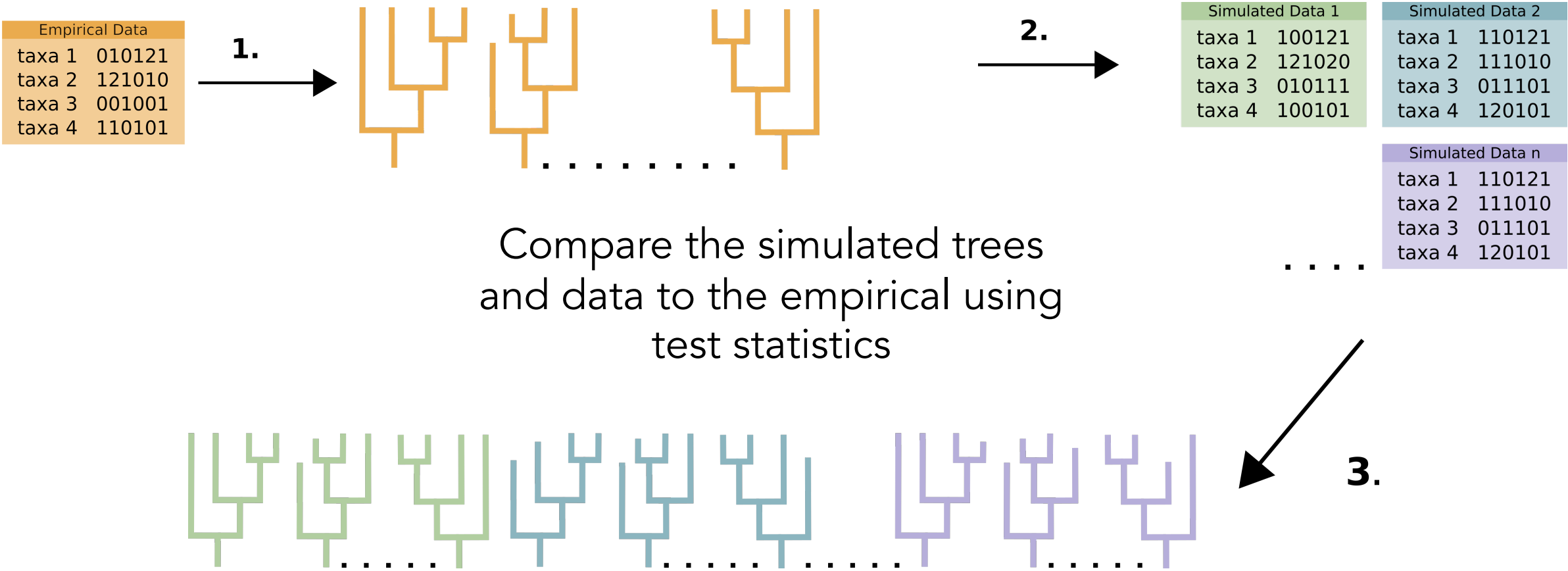
Model adequacy

Assesses whether a model is capturing the evolutionary dynamics that generated the data.

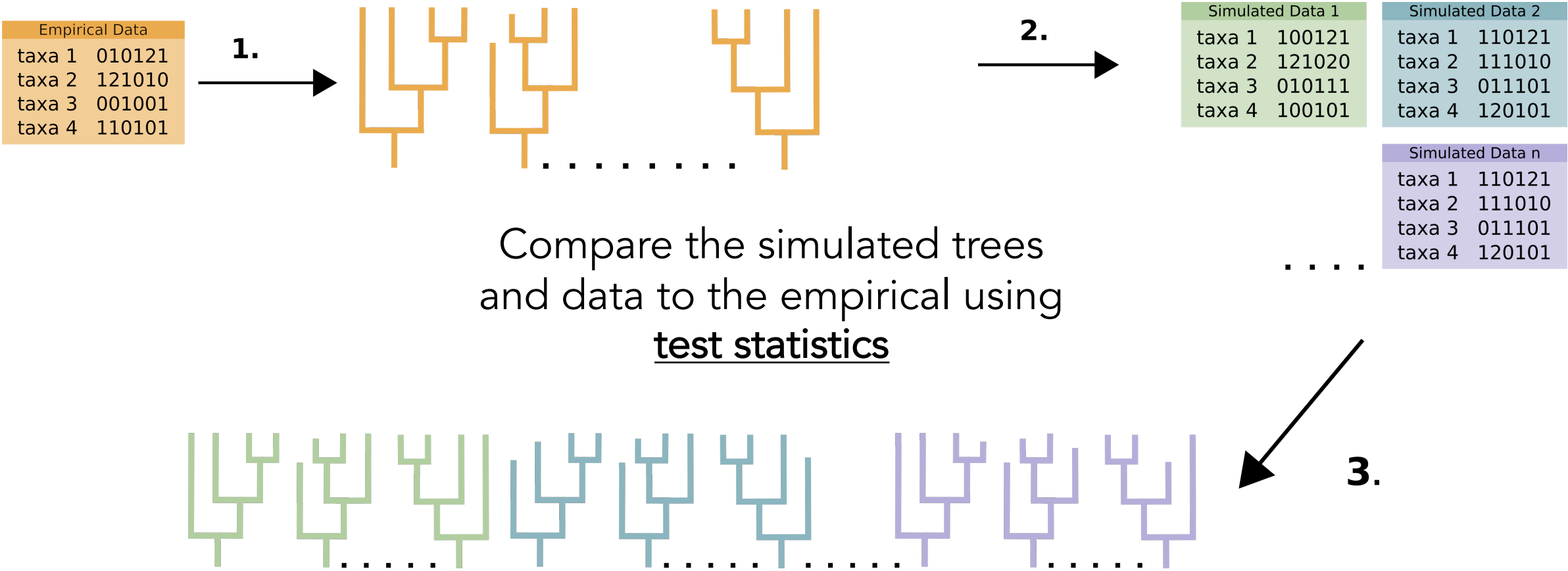
Gives the **absolute fit**

One approach is **Posterior Predictive Simulations (PPS)**

Posterior Predictive Simulations



Posterior Predictive Simulations



Test statistics

A test statistic is a **numerical summary** of data.

A value that captures the characteristic of your data.

For PPS we have 3 categories:

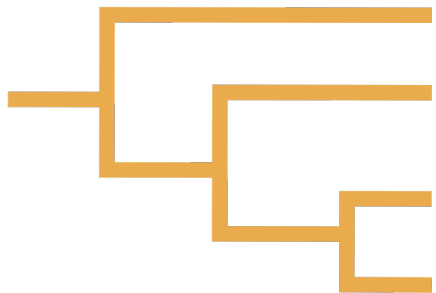
Data-based, inference-based, mixed

Test statistics: CI

Calculating consistency index

Empirical Data	
taxa 1	010121
taxa 2	121010
taxa 3	001001
taxa 4	110101

MCC summary tree



Calculate **one value** for the empirical data set

consistency index: measure of homoplasy (convergent evolution)

Simulated Data 1	
taxa 1	100121
taxa 2	121020
taxa 3	010111
taxa 4	100101

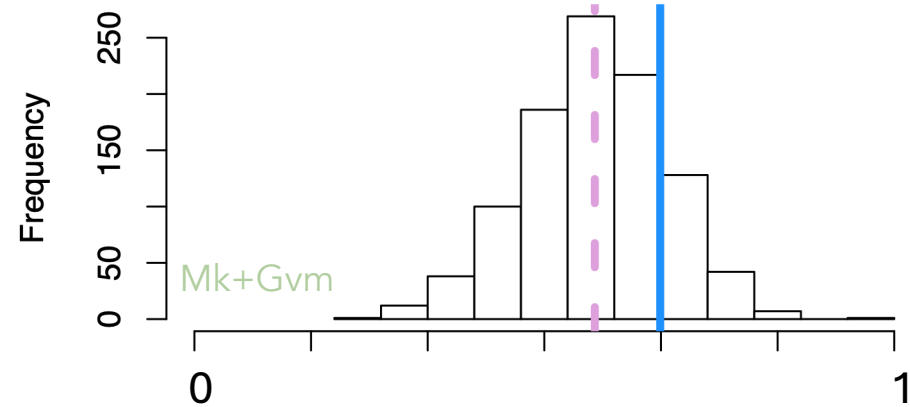
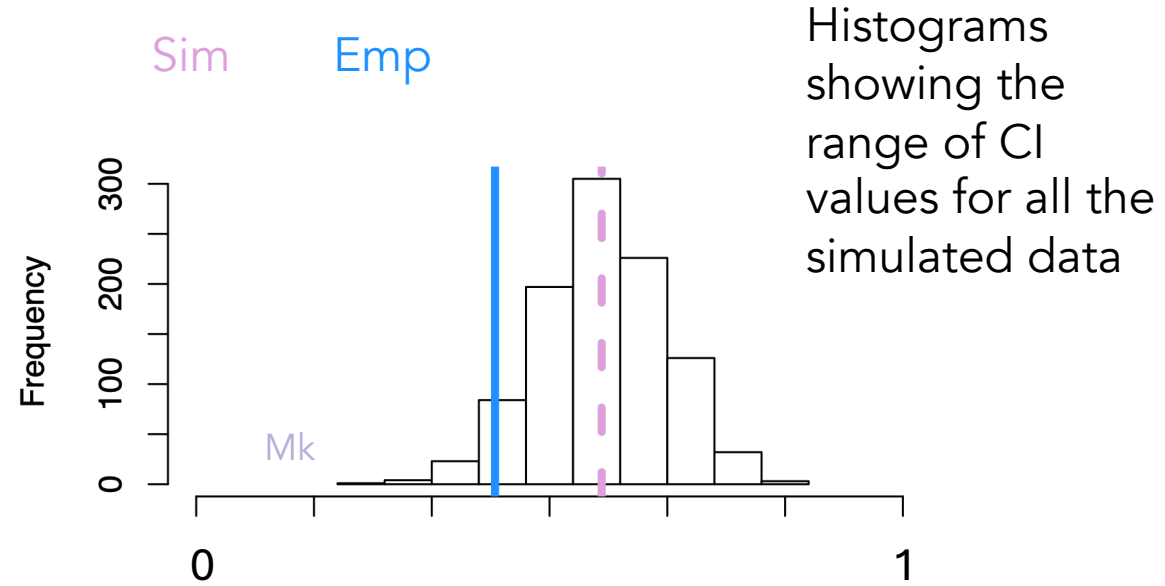
Simulated Data 2	
taxa 1	110121
taxa 2	111010
taxa 3	011101
taxa 4	120101

Simulated Data n	
taxa 1	110121
taxa 2	111010
taxa 3	011101
taxa 4	120101

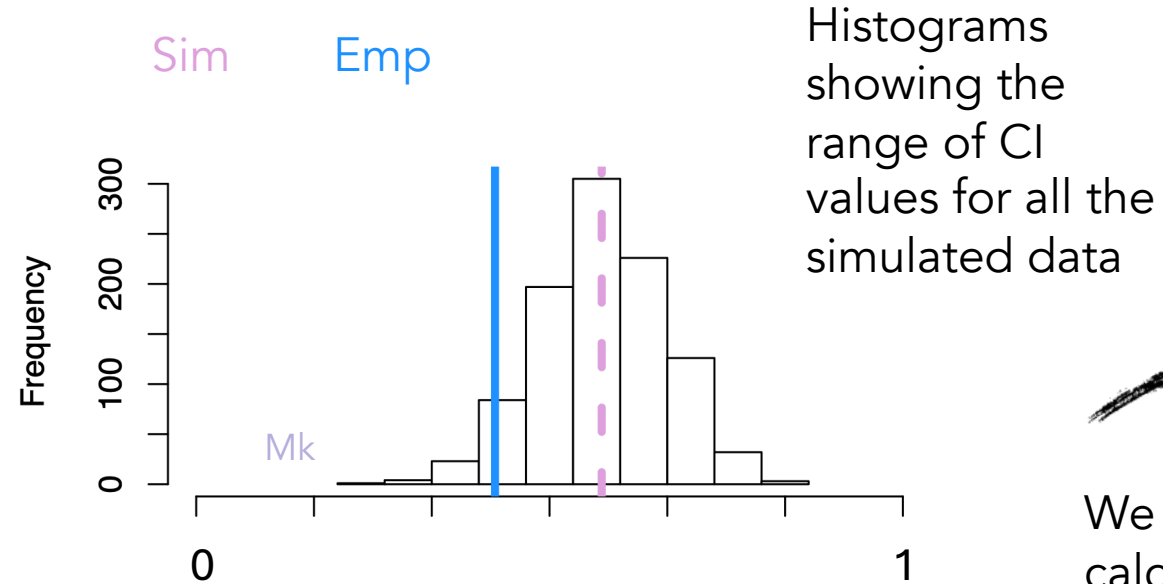
■ ■ ■ ■

Calculate a **range (500) values** using all simulated data sets

Test statistics: CI



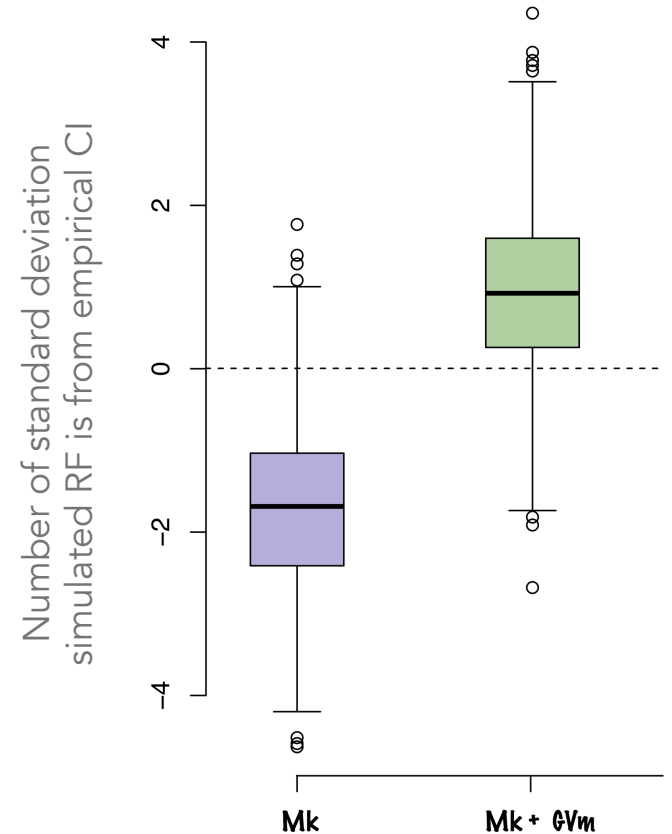
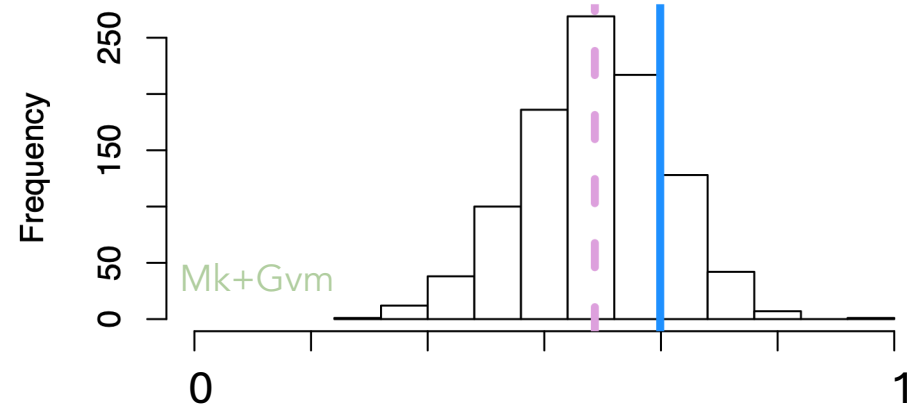
Test statistics: CI



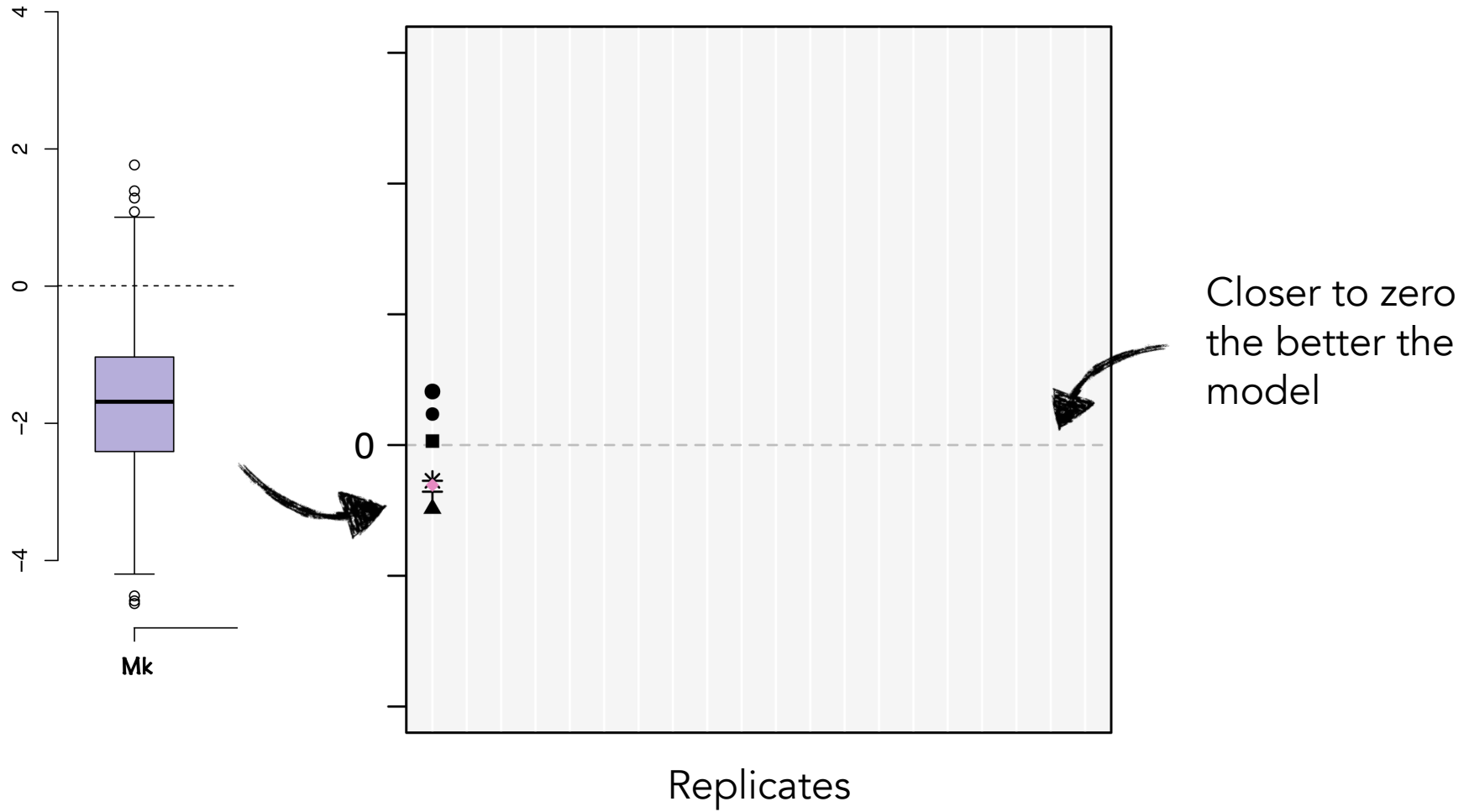
We can use this to calculate **effect sizes**

Empirical TS - SimTs

Sd(All Sim TS)



Effect sizes



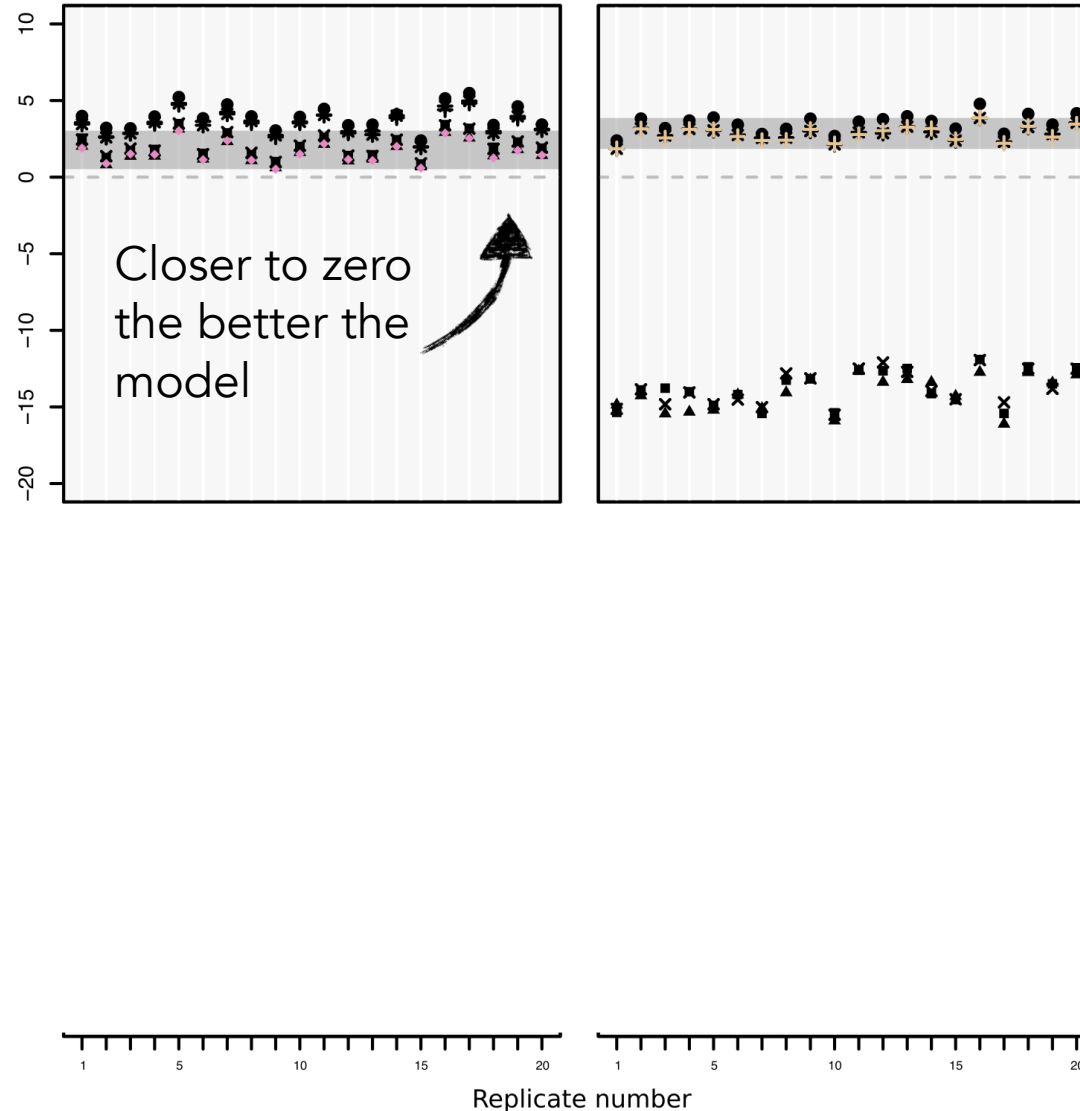
Test statistics: CI



Simulated under the MkV+G model:



Simulated under the MkVP+G model:



We do see the correct model consistently closest to zero

These test statistics **are** informative about the correct model

Empirical data sets

MkVP+G

MkVP

MkP+G

MkV+G

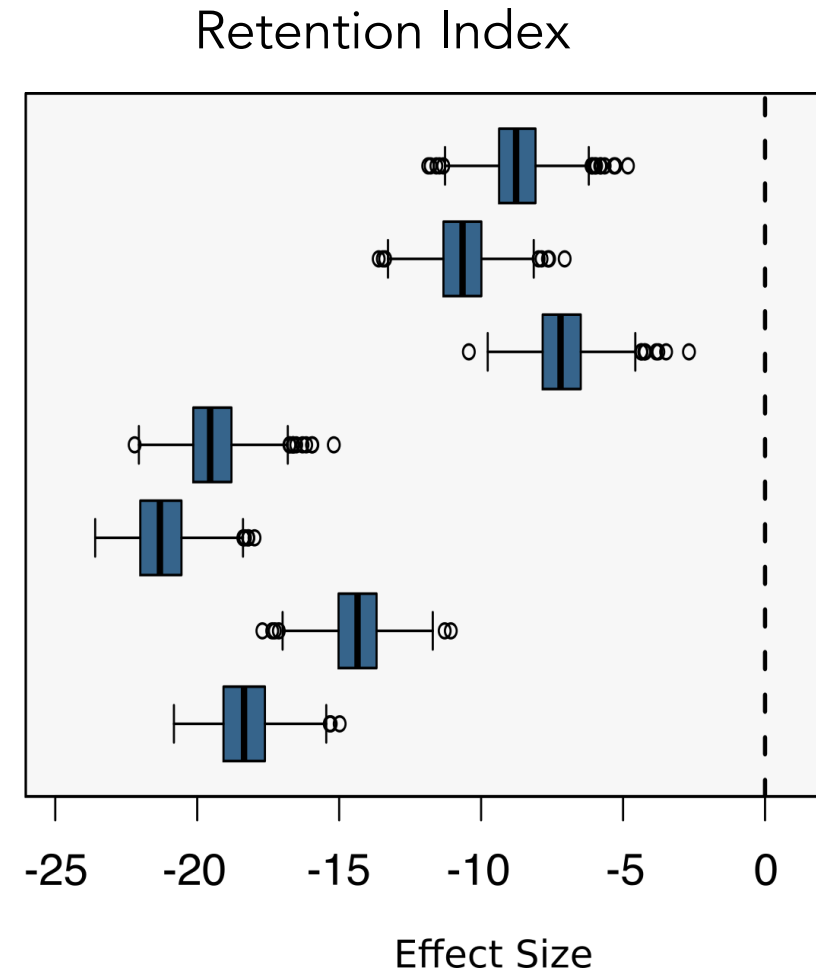
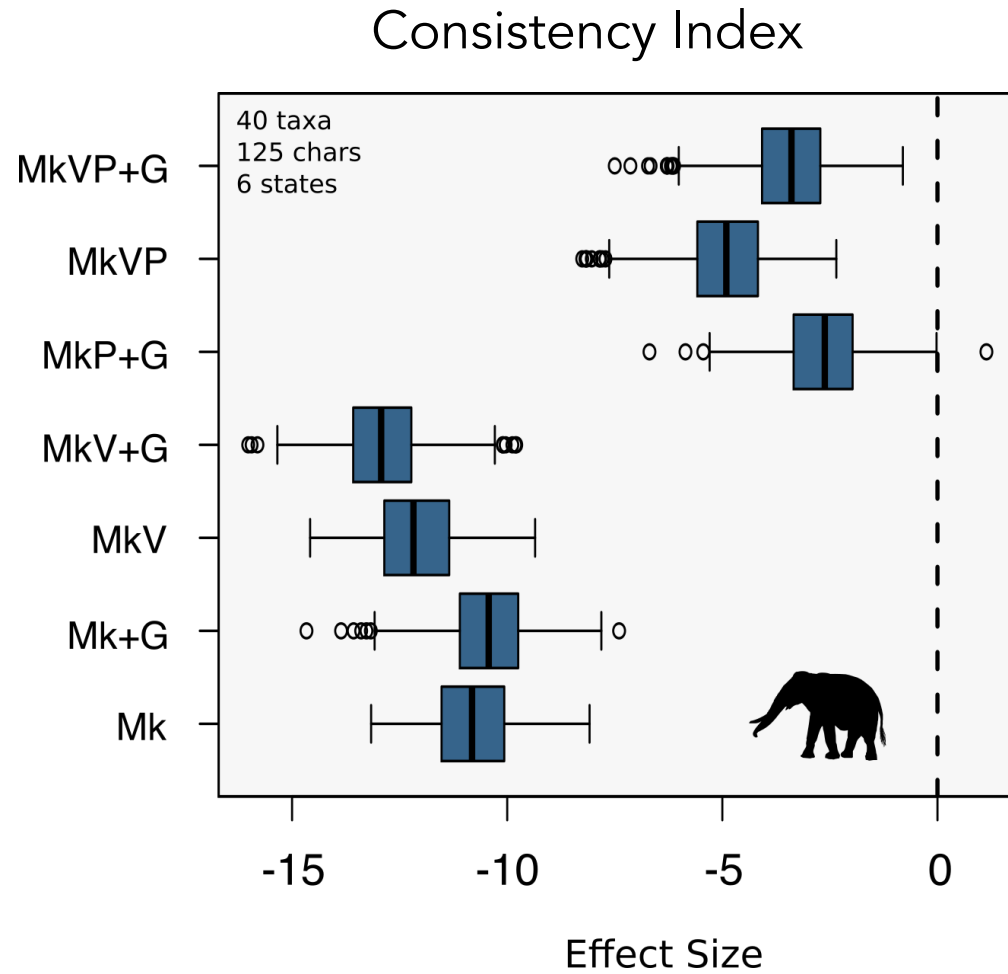
MkV

Mk+G

Mk

found 3
models
that are
dequate

Empirical data sets



No models
are
adequate

Model adequacy exercise